



**ASASIpintar UKM Programme
(Pre-University Foundation Program)**

PHYSICS I

PNAP0123

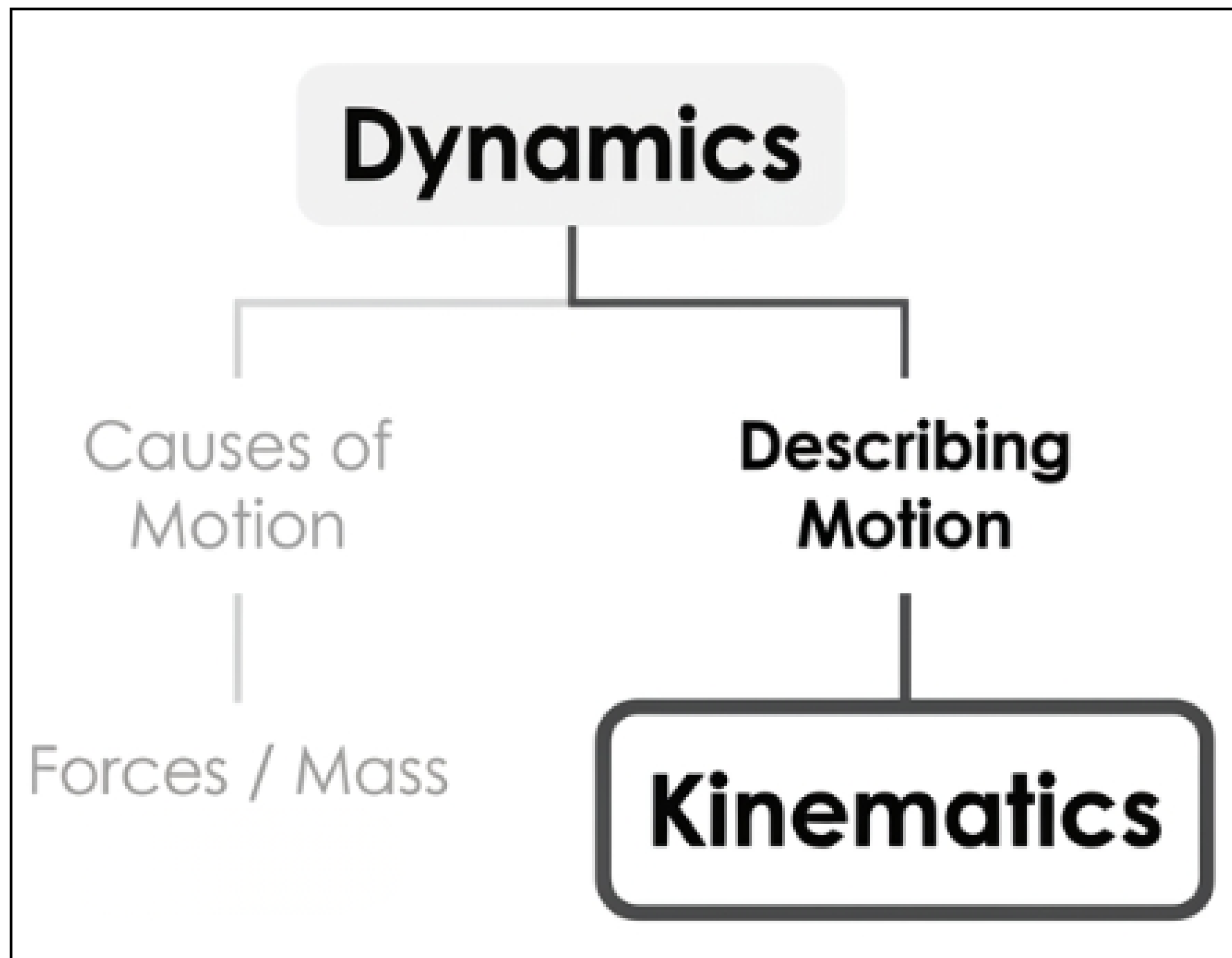
Chapter 2

Kinematics in One Dimension

Contents

1. Displacement, velocity and acceleration
2. Motion Diagrams
3. One-Dimensional Motion with Constant Acceleration
4. Freely Falling Objects

2.0 INTRODUCTION



Dynamics:

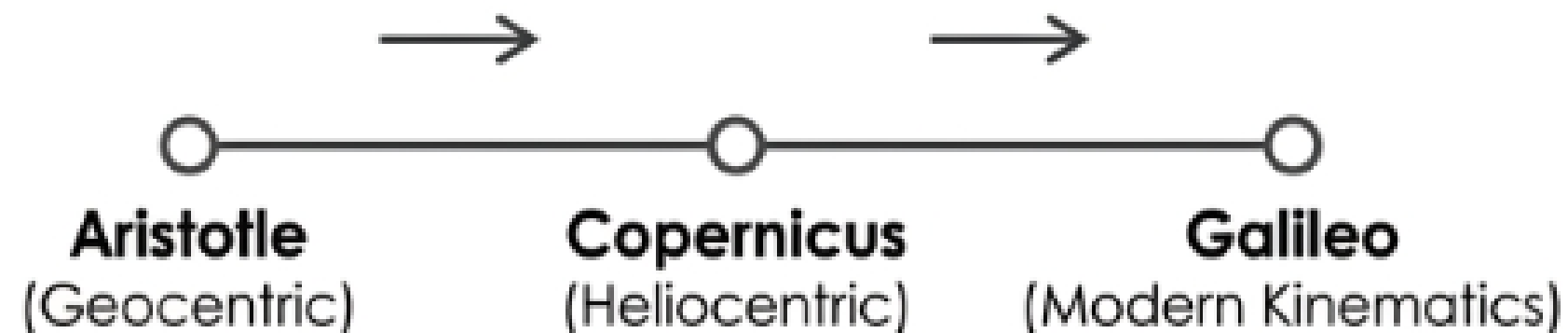
- The study of motion along with physical concepts like force and mass

Kinematics:

- The branch of dynamics describing motion without regard to what causes it

2.0 INTRODUCTION

Historically...



Early Studies: Ancient Sumerians and Egyptians first studied mechanics to understand the motions of heavenly bodies.

Geocentric Model: Greeks (300 BC–AD 300), including Aristotle and Ptolemy, placed Earth at the center of the Universe. This dominant theory stood until the 17th century.

Heliocentric Shift: Nicolaus Copernicus (1473–1543) initiated a revolution by establishing that Earth and planets revolve around the Sun.

Galileo Galilei: Initiated the modern era of physics using early telescope observations, laying the (1564–1642) quantitative foundation for Newton.

2.0 INTRODUCTION

In this chapter...

- We shall focus on the **Kinematics in one dimension** – meaning motion strictly along a straight line.
- Core Concepts:
 - We will utilize **displacement, velocity, and acceleration** to study objects moving with **constant acceleration**.

2.1 Displacement, Velocity, & Acceleration

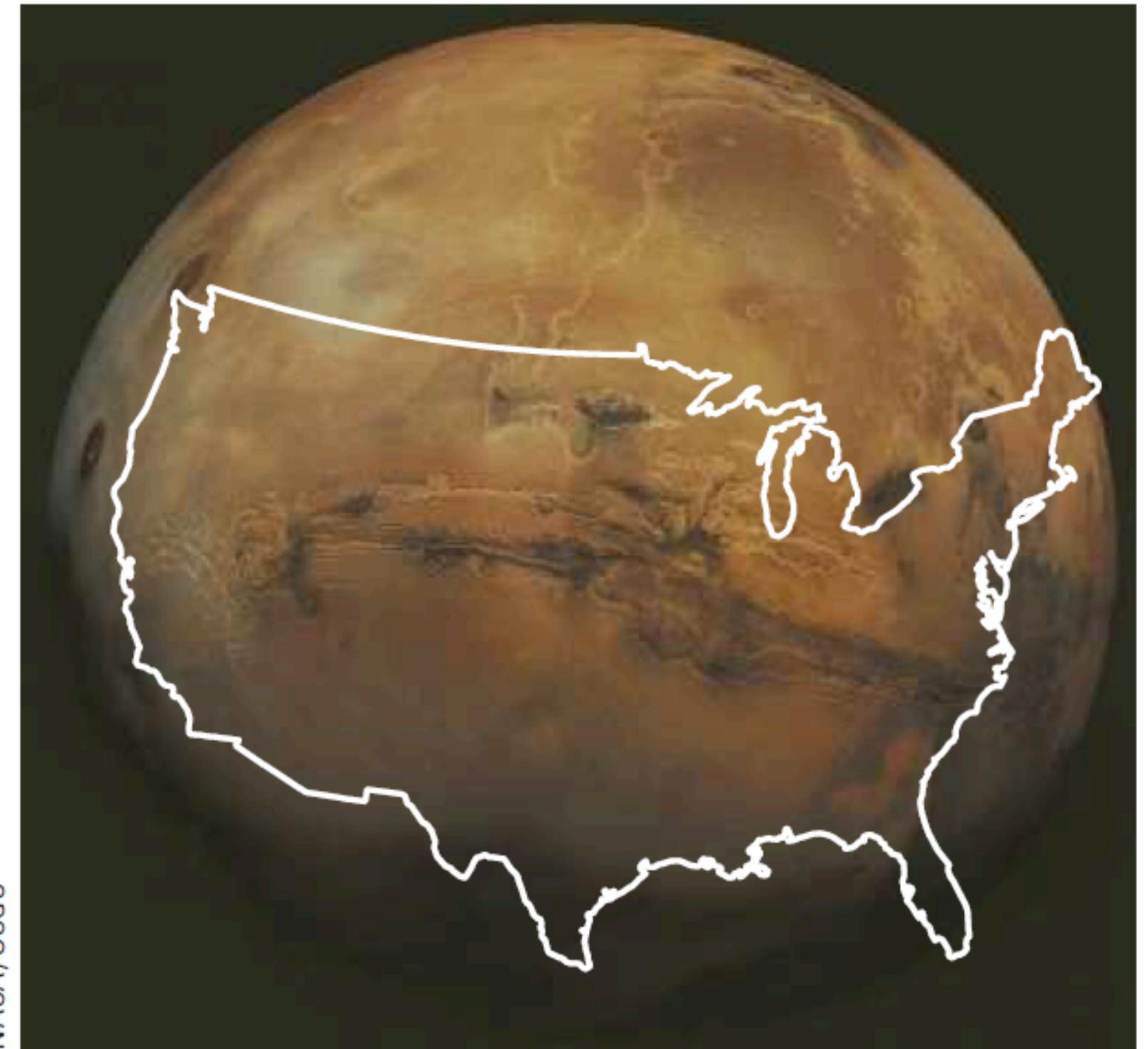
Defining a Frame of Reference

Describing motion requires a convenient coordinate system and a specified origin.

In order to do that, we to understand **frame of reference**.

Frame of Reference:

- A choice of coordinate axes defining the starting point for measuring any physical quantity.



Valles Marineris canyon without context (left) vs. the canyon with the US outline superposed as a frame of reference (above)

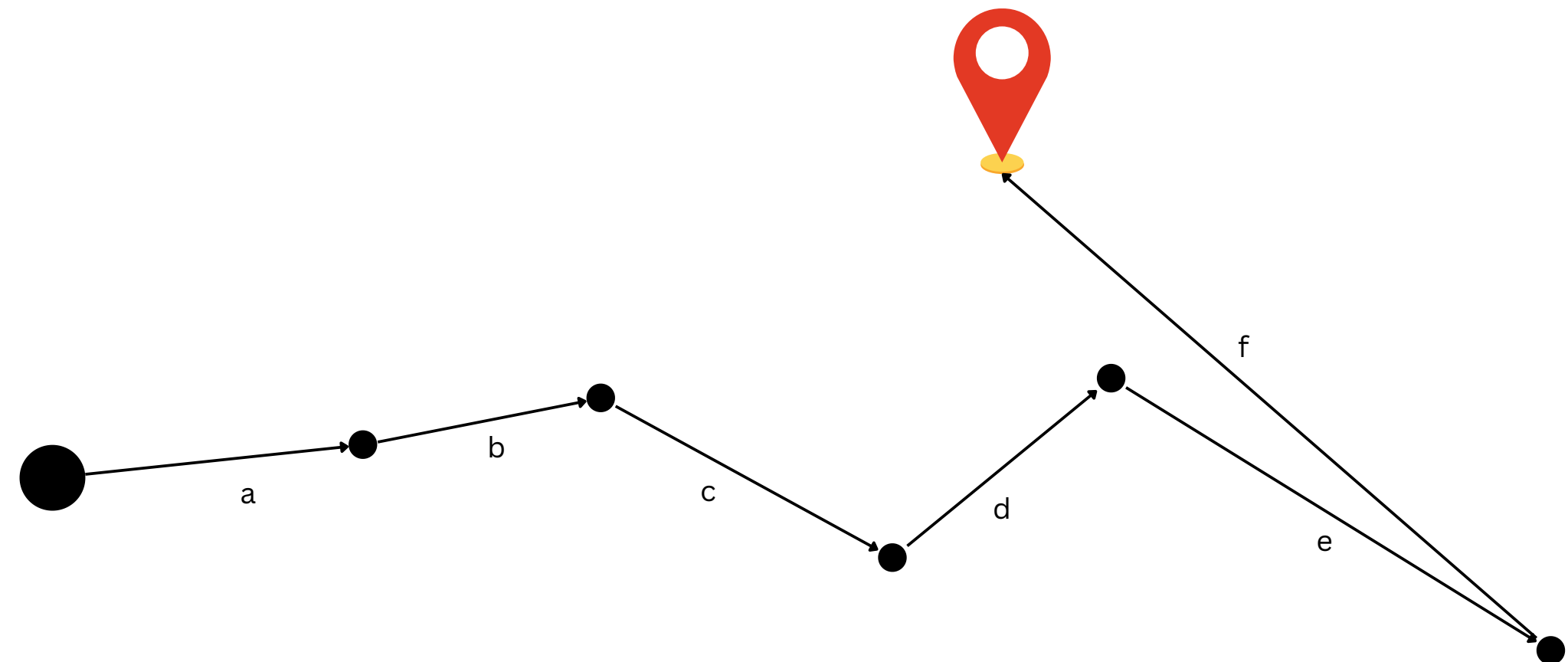
2.1 Displacement, Velocity, & Acceleration

2.1.1 Distance

- A measure of how far an object has covered during its motion.
- The total length travelled by the object (regardless of its direction).
- It is always a positive value.
- It is dependent on the path taken.



Think of it as the reading on your car's odometer. It measures the total ground covered, regardless of whether you're going forward, backward, or in circles.

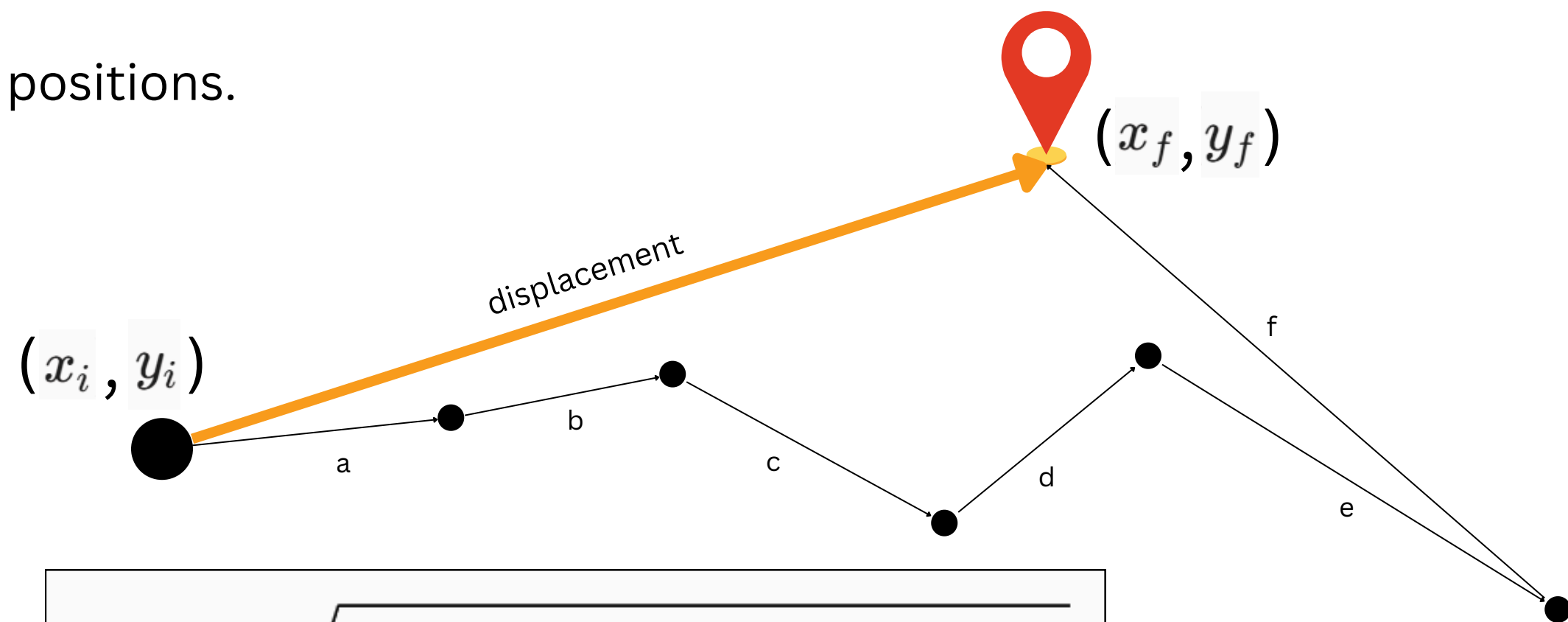


$$\text{distance} = a+b+c+d+e+f$$

2.1 Displacement, Velocity, & Acceleration

2.1.1 Displacement

- It is the shortest straight-line path between an object's starting and ending points.
- It has both a **magnitude** and a **direction**
- It can be a **positive**, **negative**, or even **zero** value. The sign indicates the direction **relative to the starting point**.
- It is independent of the path taken;
 - it only cares about the initial and final positions.

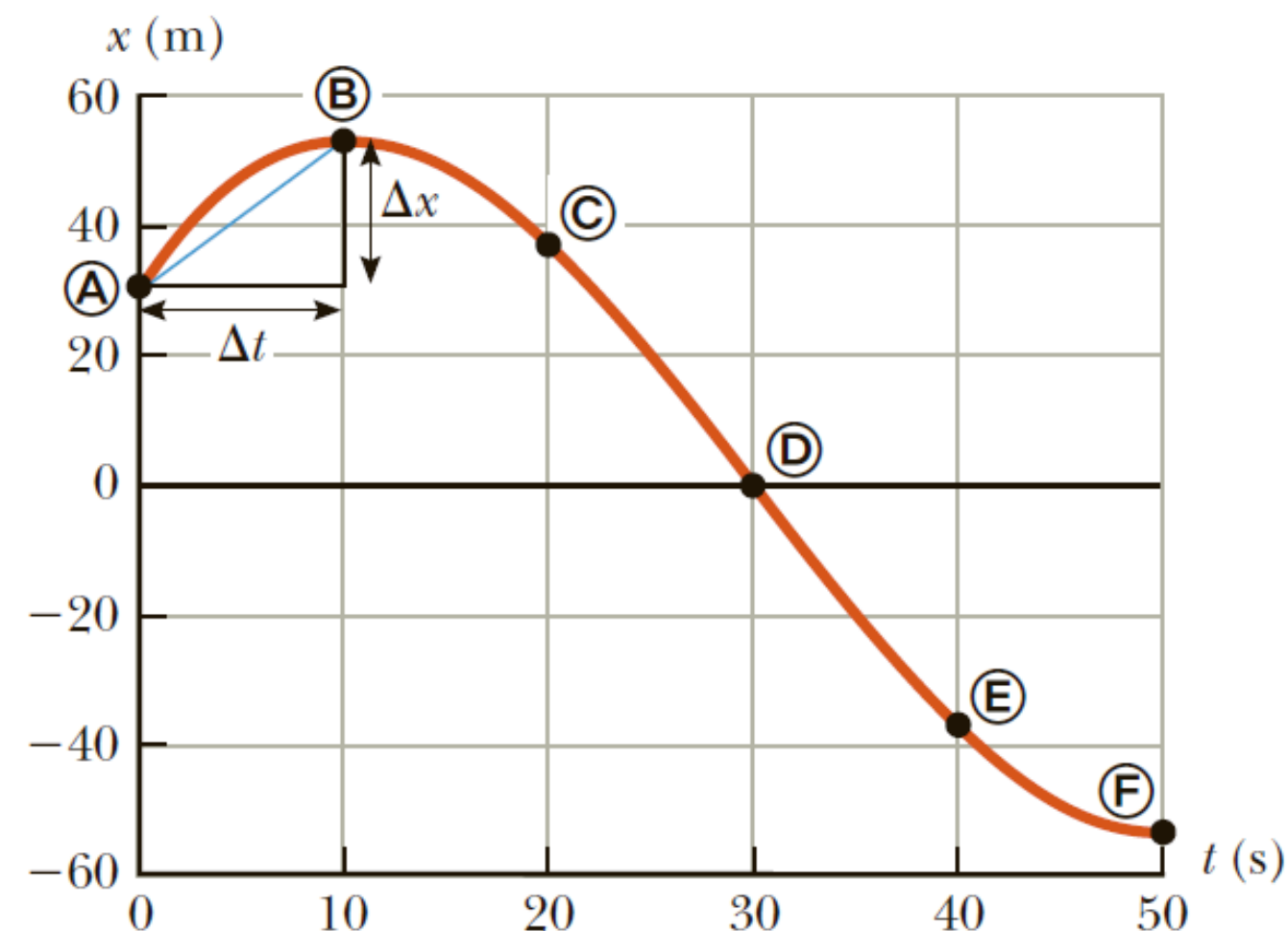
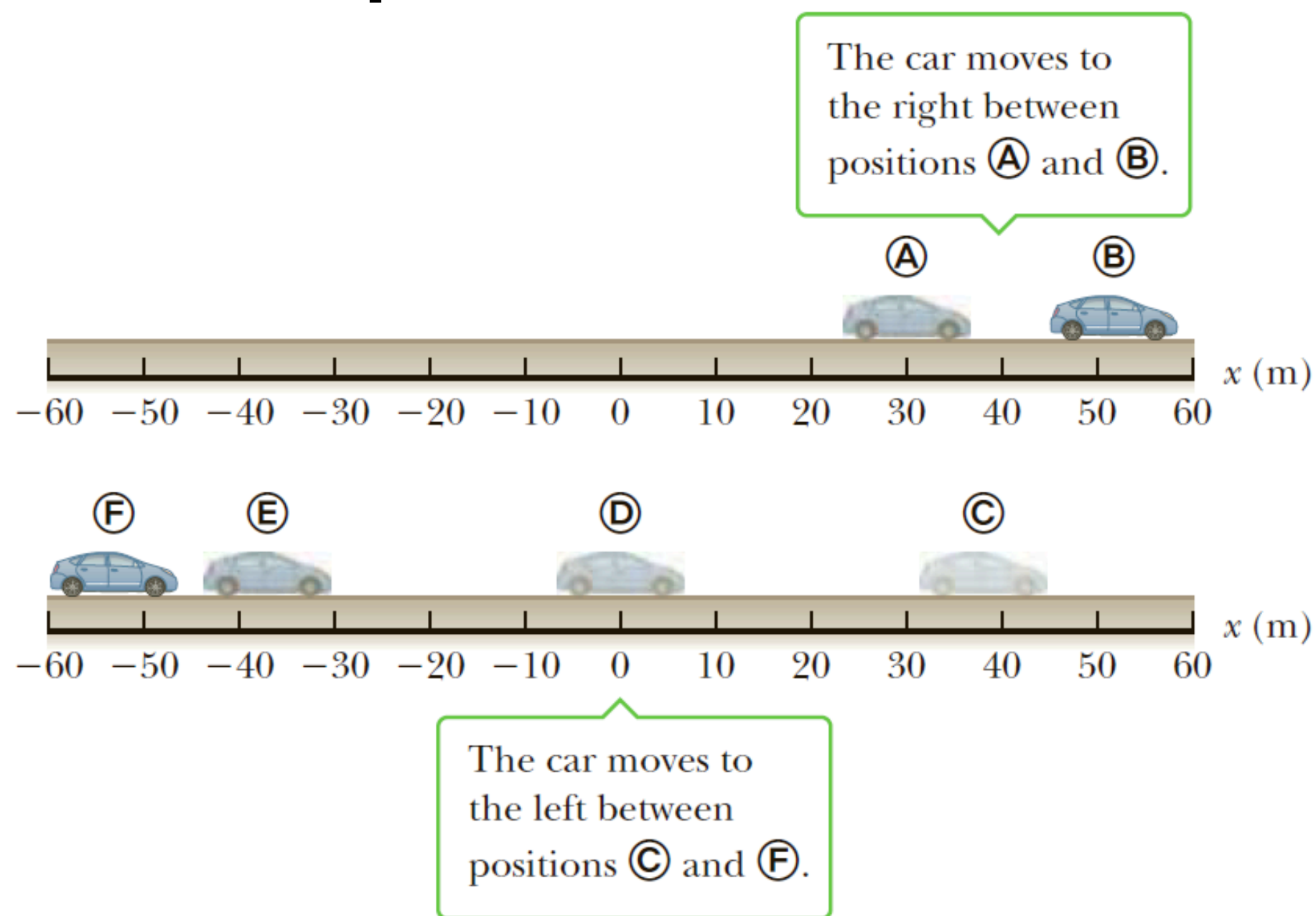


The magnitude of the displacement. It is a straight-line measure calculated via the **Pythagorean theorem**

$$\Delta s = \sqrt{(x_f - x_i)^2 + (y_f - y_i)^2}$$

2.1 Displacement, Velocity, & Acceleration

2.1.1 Displacement



A car moves back and forth along a straight line taken to be the x -axis. Because we are interested only in the car's translational motion, we can model it as a particle.

Graph of position vs. time for the motion of the "particle."

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

In everyday usage, "speed" and "velocity" are mixed up, but in physics, they are explicitly distinct

Speed

Defintion: Speed is defined as distance traveled per unit time interval.

Mathematical Equation:

$$\text{Average speed} = \frac{\text{path length}}{\text{elapsed time}}$$

Symbolic form: $v = \frac{d}{t}$ (where v is average speed, d is path length, and t is elapsed time).

SI Unit: meter per second (m/s).

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

PROBLEM A turtle and a rabbit engage in a footrace over a distance of 4.00 km. The rabbit runs 0.500 km and then stops for a 90.0-min nap. Upon awakening, he remembers the race and runs twice as fast. Finishing the course in a total time of 1.75 h, the rabbit wins the race. (a) Calculate the average speed of the rabbit. (b) What was his average speed before he stopped for a nap? Assume no detours or doubling back.

SOLUTION

(a) Find the rabbit's overall average speed.

Apply the equation for average speed:

$$\begin{aligned} \text{Average speed} &\equiv \frac{\text{path length}}{\text{elapsed time}} = \frac{4.00 \text{ km}}{1.75 \text{ h}} \\ &= 2.29 \text{ km/h} \end{aligned}$$

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

PROBLEM A turtle and a rabbit engage in a footrace over a distance of 4.00 km. The rabbit runs 0.500 km and then stops for a 90.0-min nap. Upon awakening, he remembers the race and runs twice as fast. Finishing the course in a total time of 1.75 h, the rabbit wins the race. (a) Calculate the average speed of the rabbit. (b) What was his average speed before he stopped for a nap? Assume no detours or doubling back.

SOLUTION

(b) Find the rabbit's average speed before his nap.

Sum the running times, and set the sum equal to 0.25 h:

$$t_1 + t_2 = 0.25 \text{ h}$$

Substitute $t_1 = d_1/v_1$ and $t_2 = d_2/v_2$:

$$(1) \quad \frac{d_1}{v_1} + \frac{d_2}{v_2} = 0.25 \text{ h}$$

Substitute $v_2 = 2v_1$ and the values of d_1 and d_2 into Equation (1):

$$(2) \quad \frac{0.500 \text{ km}}{v_1} + \frac{3.50 \text{ km}}{2v_1} = 0.25 \text{ h}$$

Solve Equation (2) for v_1 :

$$v_1 = 9.0 \text{ km/h}$$

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

Velocity

Definition: velocity is defined as the **rate of change** of **displacement**.

Mathematical Equation:

$$\vec{v} = \frac{\text{displacement, } \vec{s}}{\text{time, } t}$$

SI Unit: meter per second (m/s).

Sign Convention: Unlike speed, velocity can be positive or negative based on the sign of displacement.
(the time interval, ***t*** is always positive)

- Positive (+) = Motion to the right.
- Negative (—) = Motion to the left.

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

Velocity

Uniform velocity

Velocity always constant (for object moves with uniform velocity, its instantaneous velocity equals to the average velocity at any time)

Average Velocity

the rate of change of displacement over a finite interval of time.

$$\bar{v} \equiv \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

Instantaneous velocity

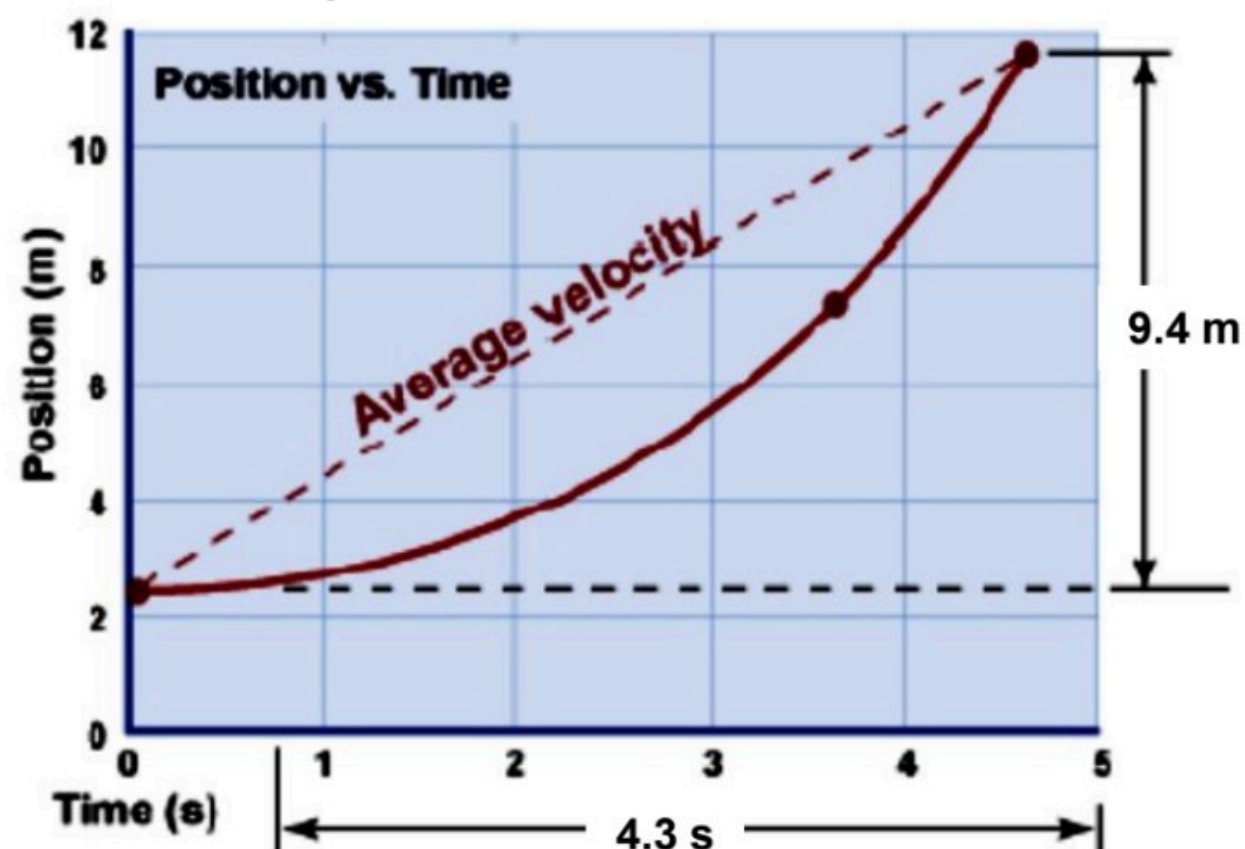
velocity at a specified position or instant of time along the path of motion.

$$v \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$$

2.1 Displacement, Velocity, & Acceleration

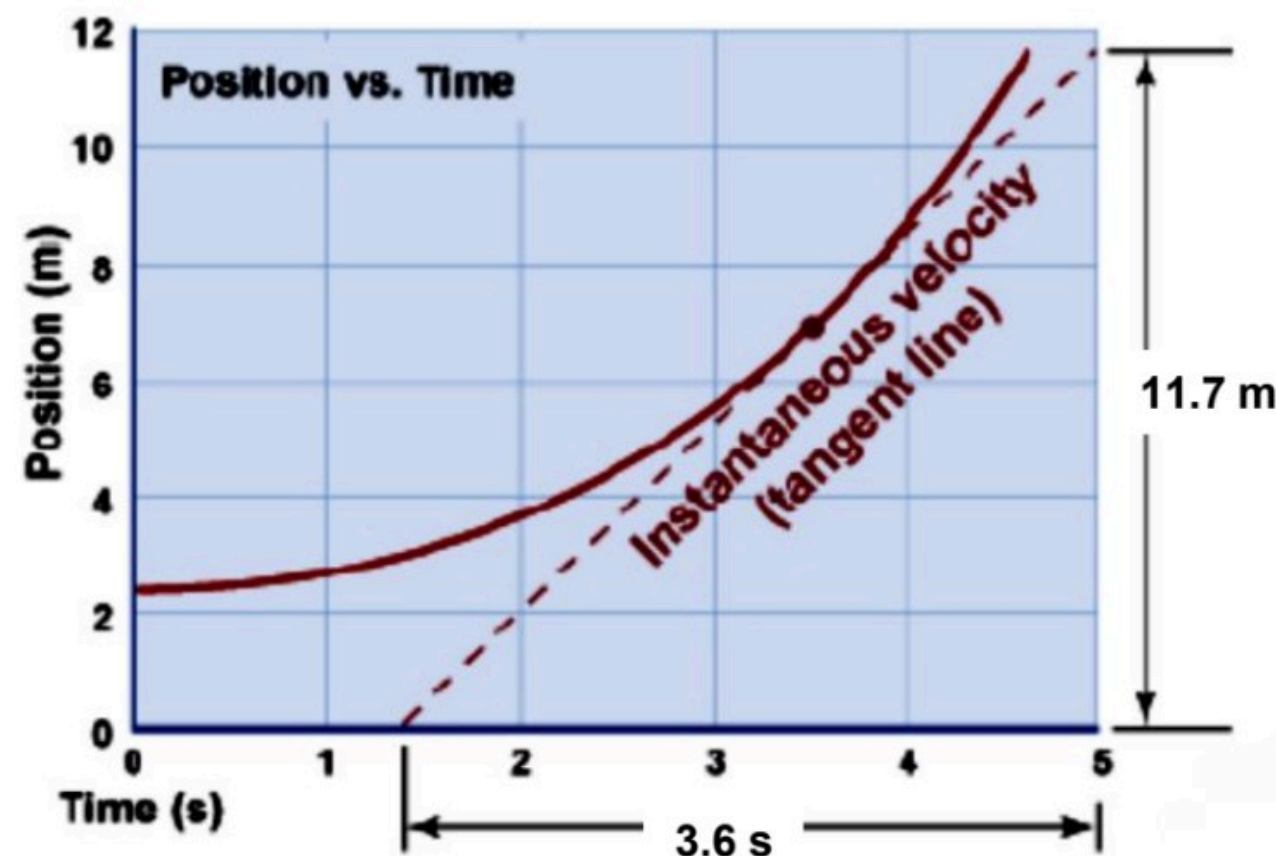
2.1.2 Speed & Velocity

Velocity



Average velocity
defined by the change between two points

$$\vec{v}_{av} = \frac{\Delta \vec{s}}{\Delta t} = \frac{9.4 \text{ m}}{4.3 \text{ s}} = 2.04 \text{ m s}^{-1}$$



Instantaneous velocity
Velocity at a point - the slope (gradient) of the tangent line

$$\vec{v} = \frac{d\vec{s}}{dt} = \frac{11.7 \text{ m}}{3.6 \text{ s}} = 3.25 \text{ m s}^{-1}$$

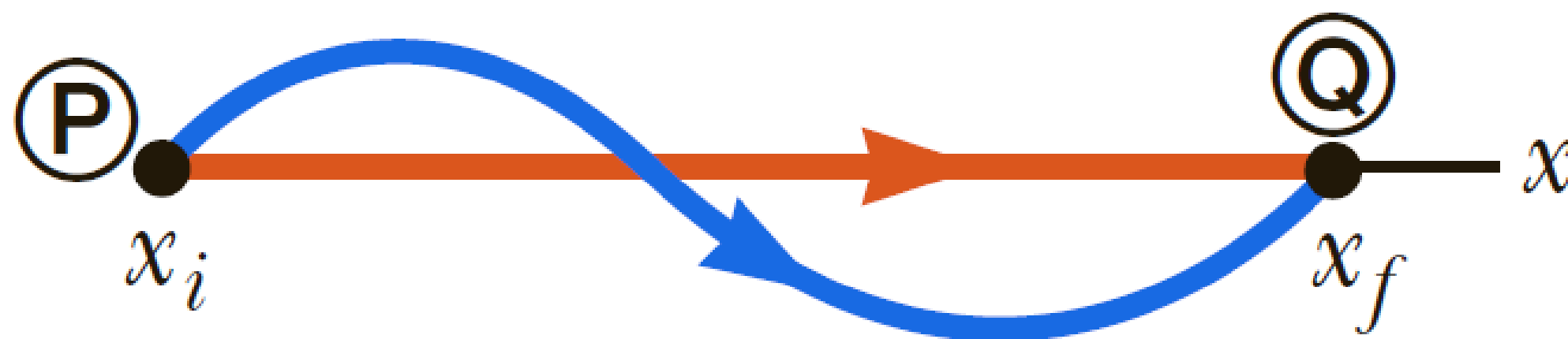
If the body speeds up or slows down during the displacement, then the average velocity is not the same as the velocity at a given instant of time during the motion

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

Velocity

To further illustrate the distinction between speed and velocity, let imagine we're watching a drag race from a stationary blimp



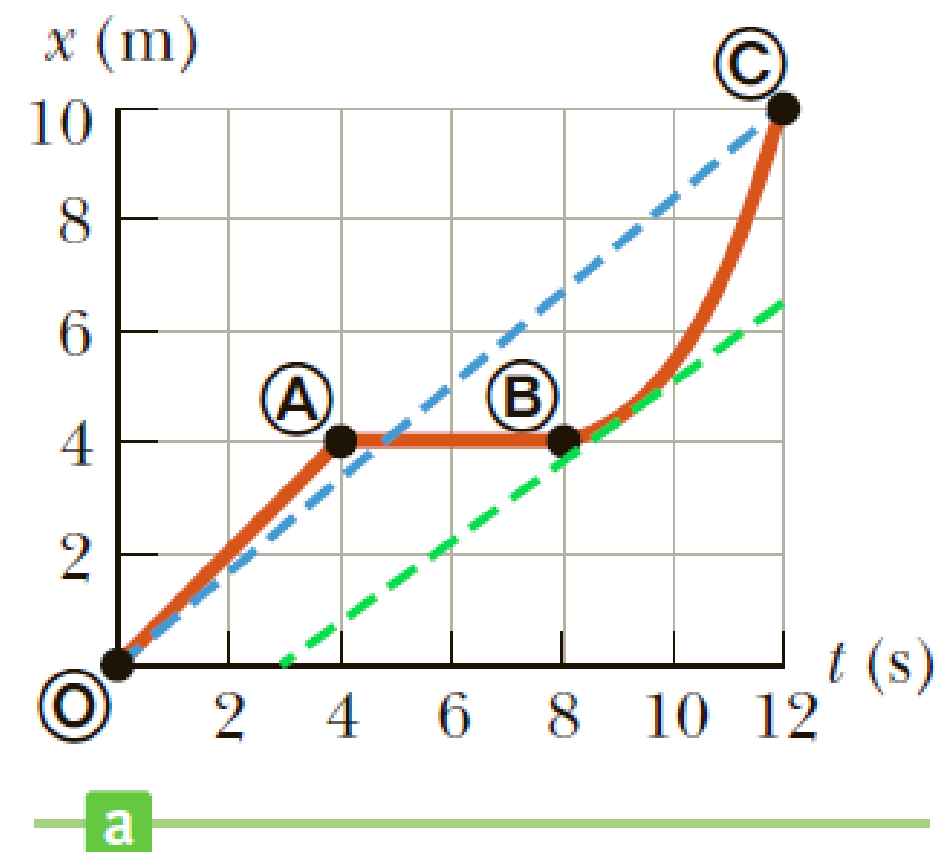
$$\bar{v} \equiv \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

- In one run, we see a car follow the straight-line path from P to Q during the time interval Δt .
- In a second run a car follows the curved path during the same interval.
- From the definition, the **two cars had the same average velocity** because they had the same displacement during the same time interval.
- The **car taking the curved route**, however, traveled a greater path length and had the **higher average speed**

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

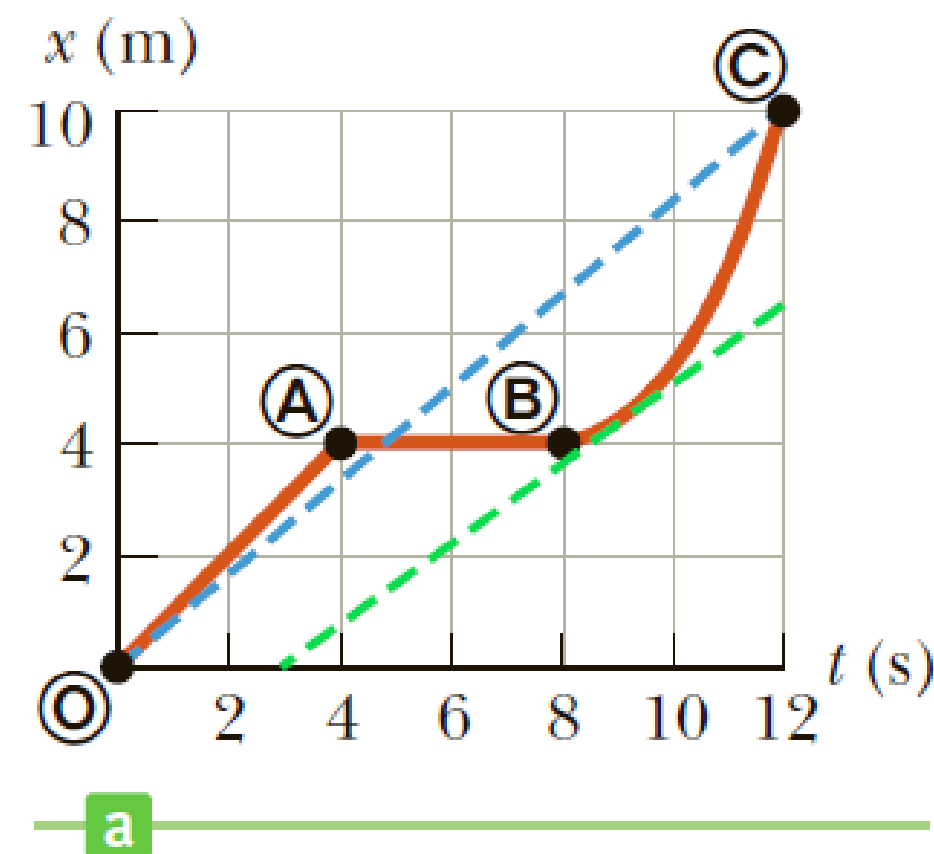
PROBLEM A train moves slowly along a straight portion of track according to the graph of position vs. time in Figure 2.7a. Find (a) the average velocity for the total trip, (b) the average velocity during the first 4.00 s of motion, (c) the average velocity during the next 4.00 s of motion, (d) the instantaneous velocity at $t = 2.00$ s, and (e) the instantaneous velocity at $t = 9.00$ s.



2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

PROBLEM A train moves slowly along a straight portion of track according to the graph of position vs. time in Figure 2.7a. Find (a) the average velocity for the total trip, (b) the average velocity during the first 4.00 s of motion, (c) the average velocity during the next 4.00 s of motion, (d) the instantaneous velocity at $t = 2.00$ s, and (e) the instantaneous velocity at $t = 9.00$ s.



SOLUTION

(a) Find the average velocity from © to ©.

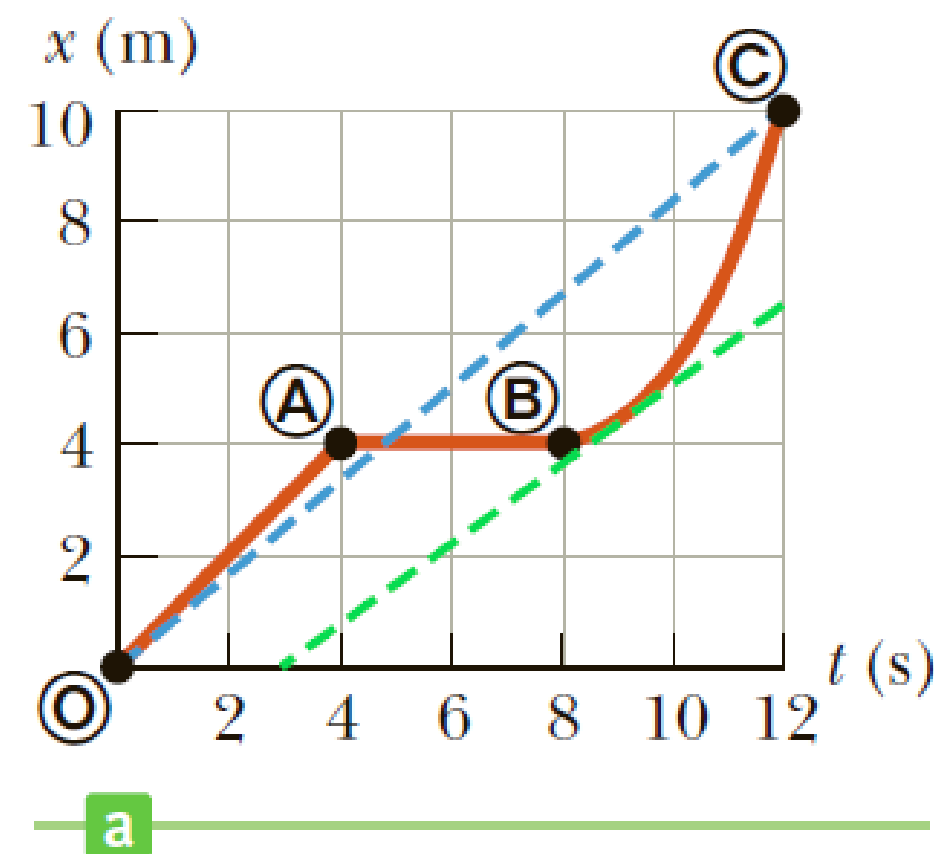
Calculate the slope of the dashed blue line:

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{10.0 \text{ m}}{12.0 \text{ s}} = +0.833 \text{ m/s}$$

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

PROBLEM A train moves slowly along a straight portion of track according to the graph of position vs. time in Figure 2.7a. Find (a) the average velocity for the total trip, (b) the average velocity during the first 4.00 s of motion, (c) the average velocity during the next 4.00 s of motion, (d) the instantaneous velocity at $t = 2.00$ s, and (e) the instantaneous velocity at $t = 9.00$ s.



SOLUTION

(b) Find the average velocity during the first 4 seconds of the train's motion.

Again, find the slope:

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{4.00 \text{ m}}{4.00 \text{ s}} = +1.00 \text{ m/s}$$

(c) Find the average velocity during the next 4 seconds.

Here, there is no change in position as the train moves from A to B, so the displacement Δx is zero:

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{0 \text{ m}}{4.00 \text{ s}} = 0 \text{ m/s}$$

2.1 Displacement, Velocity, & Acceleration

2.1.2 Speed & Velocity

PROBLEM A train moves slowly along a straight portion of track according to the graph of position vs. time in Figure 2.7a. Find (a) the average velocity for the total trip, (b) the average velocity during the first 4.00 s of motion, (c) the average velocity during the next 4.00 s of motion, (d) the instantaneous velocity at $t = 2.00$ s, and (e) the instantaneous velocity at $t = 9.00$ s.

SOLUTION

(d) Find the instantaneous velocity at $t = 2.00$ s.

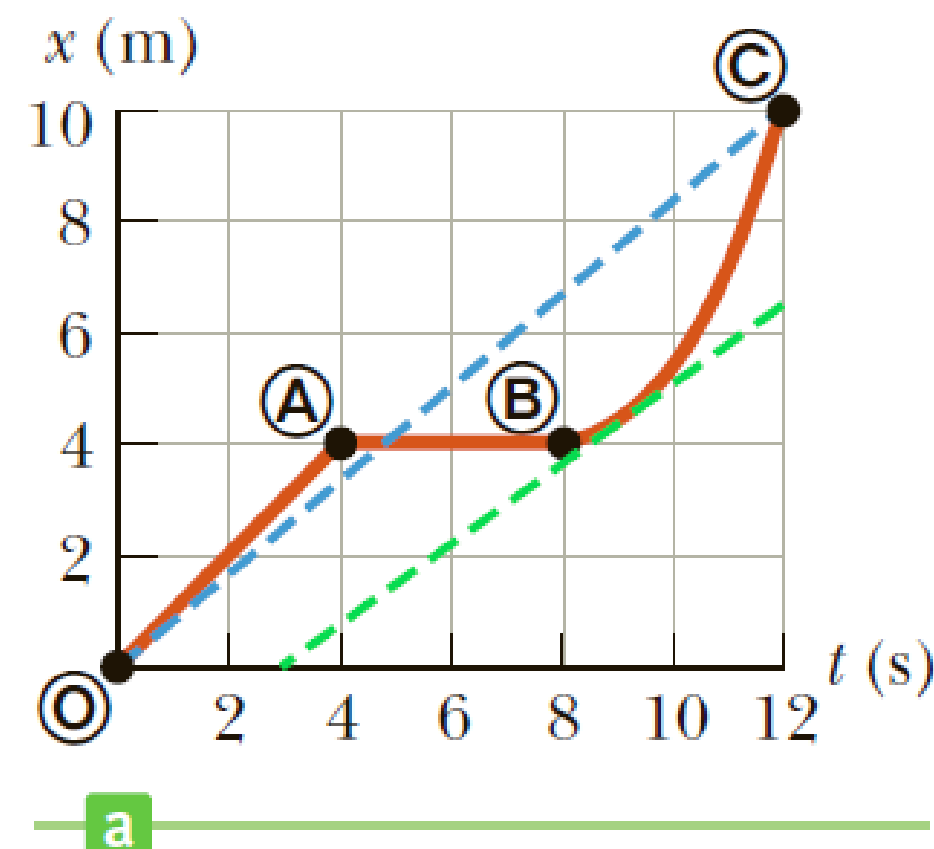
This is the same as the average velocity found in (b), because the graph is a straight line:

$$v = 1.00 \text{ m/s}$$

(e) Find the instantaneous velocity at $t = 9.00$ s.

The tangent line appears to intercept the x -axis at (3.0 s, 0 m) and graze the curve at (9.0 s, 4.5 m). The instantaneous velocity at $t = 9.00$ s equals the slope of the tangent line through these points:

$$v = \frac{\Delta x}{\Delta t} = \frac{4.5 \text{ m} - 0 \text{ m}}{9.0 \text{ s} - 3.0 \text{ s}} = 0.75 \text{ m/s}$$



2.1 Displacement, Velocity, & Acceleration

2.1.3 Acceleration

In general, acceleration is the changing of an object's velocity over time.

Definition: Acceleration is defined as the rate of change of velocity

$$a = \frac{\Delta v}{\Delta t}$$

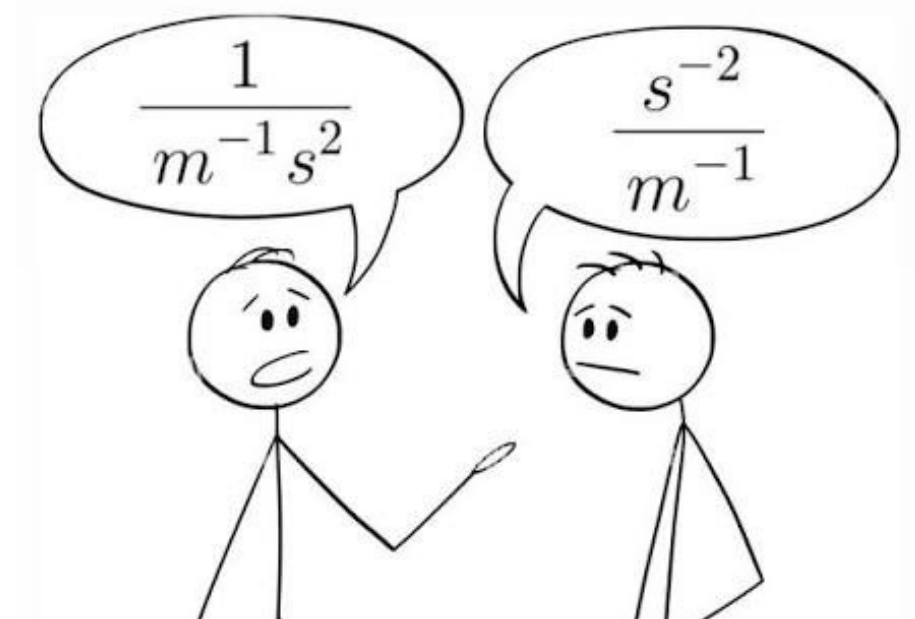
S.I. unit is m s^{-2}

It is a vector quantity

An acceleration may due to change in:

- speed (magnitude),
- direction
- both speed or direction

Physics Classmates be Like
acceleration



2.1 Displacement, Velocity, & Acceleration

2.1.3 Acceleration

Average acceleration,

Defintion: Average acceleration defined as change in velocity divided by the time taken to make the change.

$$\vec{a}_{av} = \frac{\text{change of velocity}}{\text{time interval}} = \frac{v_2 - v_1}{t_2 - t_1}$$
$$a_{av} = \frac{\Delta v}{\Delta t}$$

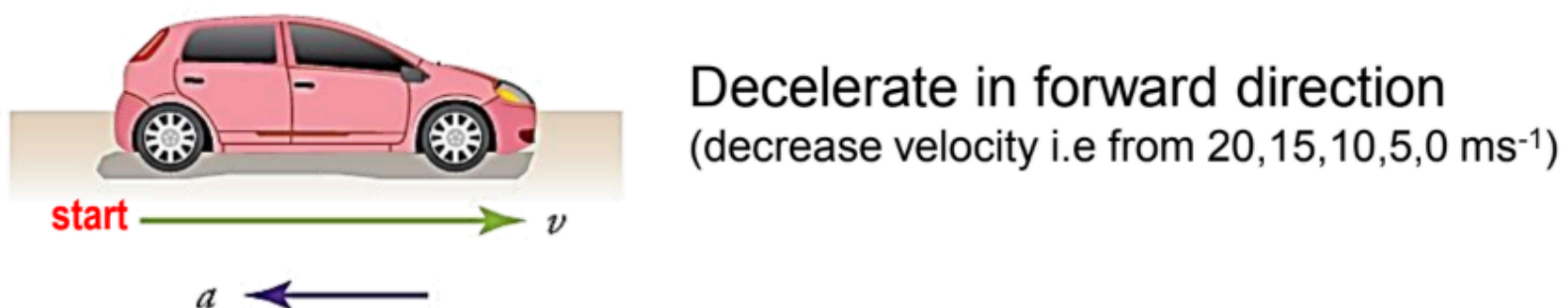
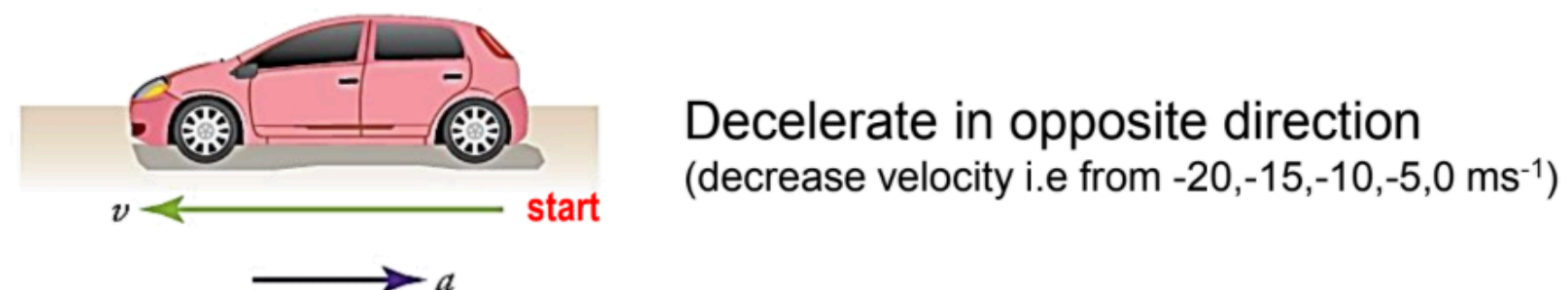
Direction of average acceleration:

- **Same direction** as direction of motion if an **object accelerates** (increase in velocity)
- **Opposite direction** to direction of motion if the **object decelerates** (decrease in velocity)

2.1 Displacement, Velocity, & Acceleration

2.1.3 Acceleration

Average acceleration



2.1 Displacement, Velocity, & Acceleration

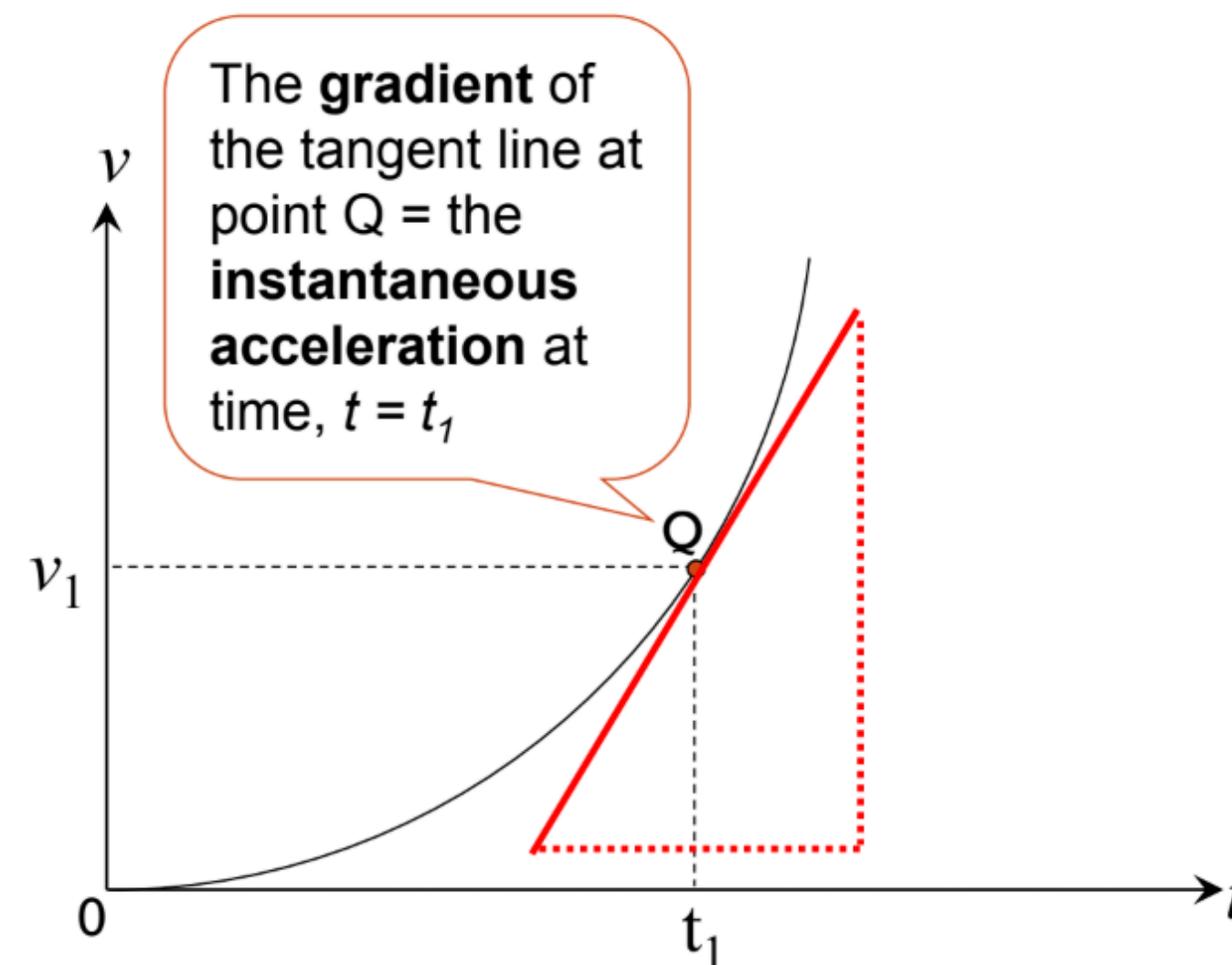
2.1.3 Acceleration

Instantaneous Acceleration

Defintion: Average acceleration is defined as the acceleration at particular instant of time or position.

Mathematical Equation:

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$



2.1 Displacement, Velocity, & Acceleration

2.1.3 Acceleration

Uniform Acceleration

An object moves in a uniform acceleration when magnitude of velocity changes at a constant rate and along fixed direction.

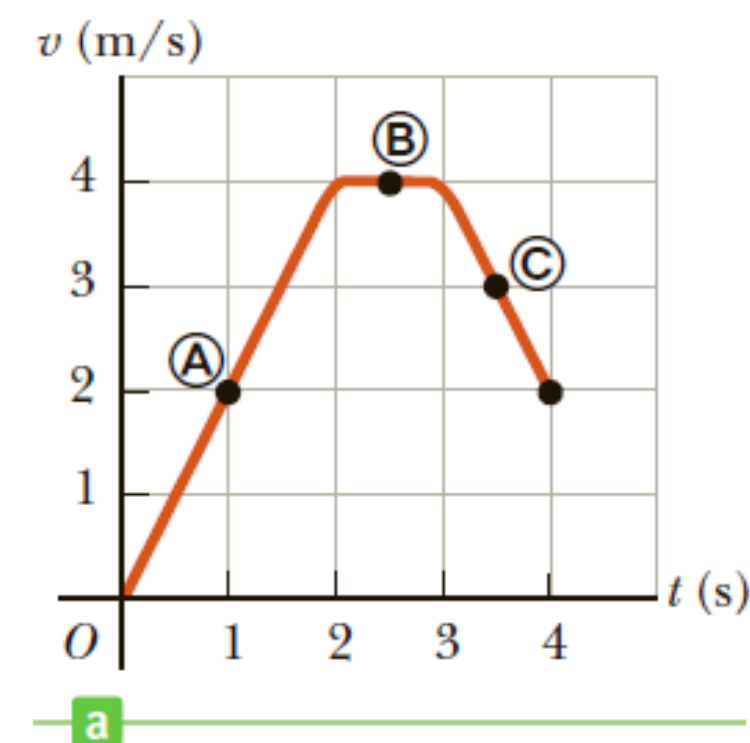
$$\frac{dv}{dt} = \text{constant}$$

For object moves with uniform acceleration its instantaneous acceleration equals to the average acceleration at any time.

2.1 Displacement, Velocity, & Acceleration

2.1.3 Acceleration

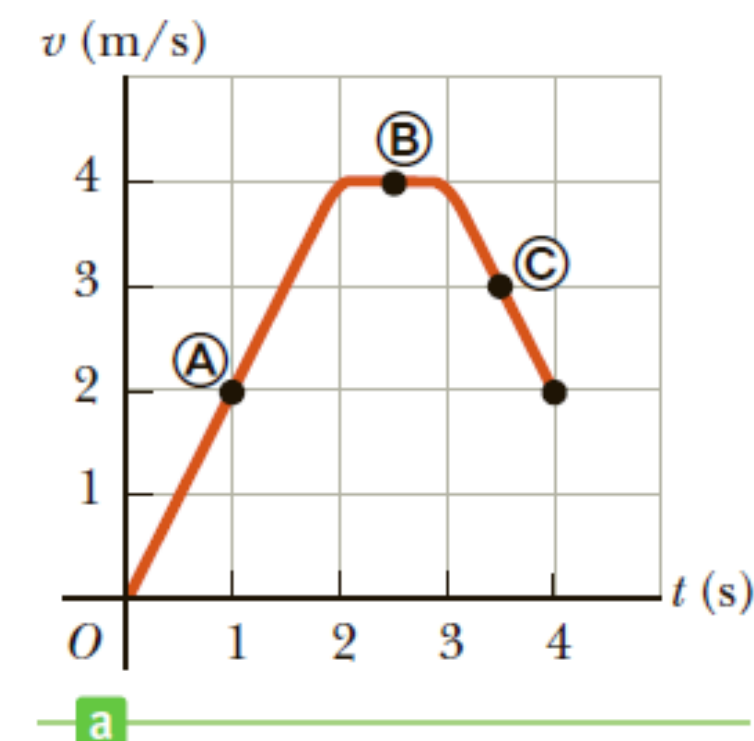
PROBLEM A baseball player moves in a straight-line path in order to catch a fly ball hit to the outfield. His velocity as a function of time is shown in Figure 2.11a. Find his instantaneous acceleration at points **A**, **B**, and **C**.



2.1 Displacement, Velocity, & Acceleration

2.1.3 Acceleration

PROBLEM A baseball player moves in a straight-line path in order to catch a fly ball hit to the outfield. His velocity as a function of time is shown in Figure 2.11a. Find his instantaneous acceleration at points **A**, **B**, and **C**.



SOLUTION

Acceleration at **A**.

The acceleration at **A** equals the slope of the line connecting the points (0 s, 0 m/s) and (2.0 s, 4.0 m/s):

$$a = \frac{\Delta v}{\Delta t} = \frac{4.0 \text{ m/s} - 0}{2.0 \text{ s} - 0} = +2.0 \text{ m/s}^2$$

Acceleration at **B**.

$\Delta v = 0$, because the segment is horizontal:

$$a = \frac{\Delta v}{\Delta t} = \frac{4.0 \text{ m/s} - 4.0 \text{ m/s}}{3.0 \text{ s} - 2.0 \text{ s}} = 0 \text{ m/s}^2$$

Acceleration at **C**.

The acceleration at **C** equals the slope of the line connecting the points (3.0 s, 4.0 m/s) and (4.0 s, 2.0 m/s):

$$a = \frac{\Delta v}{\Delta t} = \frac{2.0 \text{ m/s} - 4.0 \text{ m/s}}{4.0 \text{ s} - 3.0 \text{ s}} = -2.0 \text{ m/s}^2$$

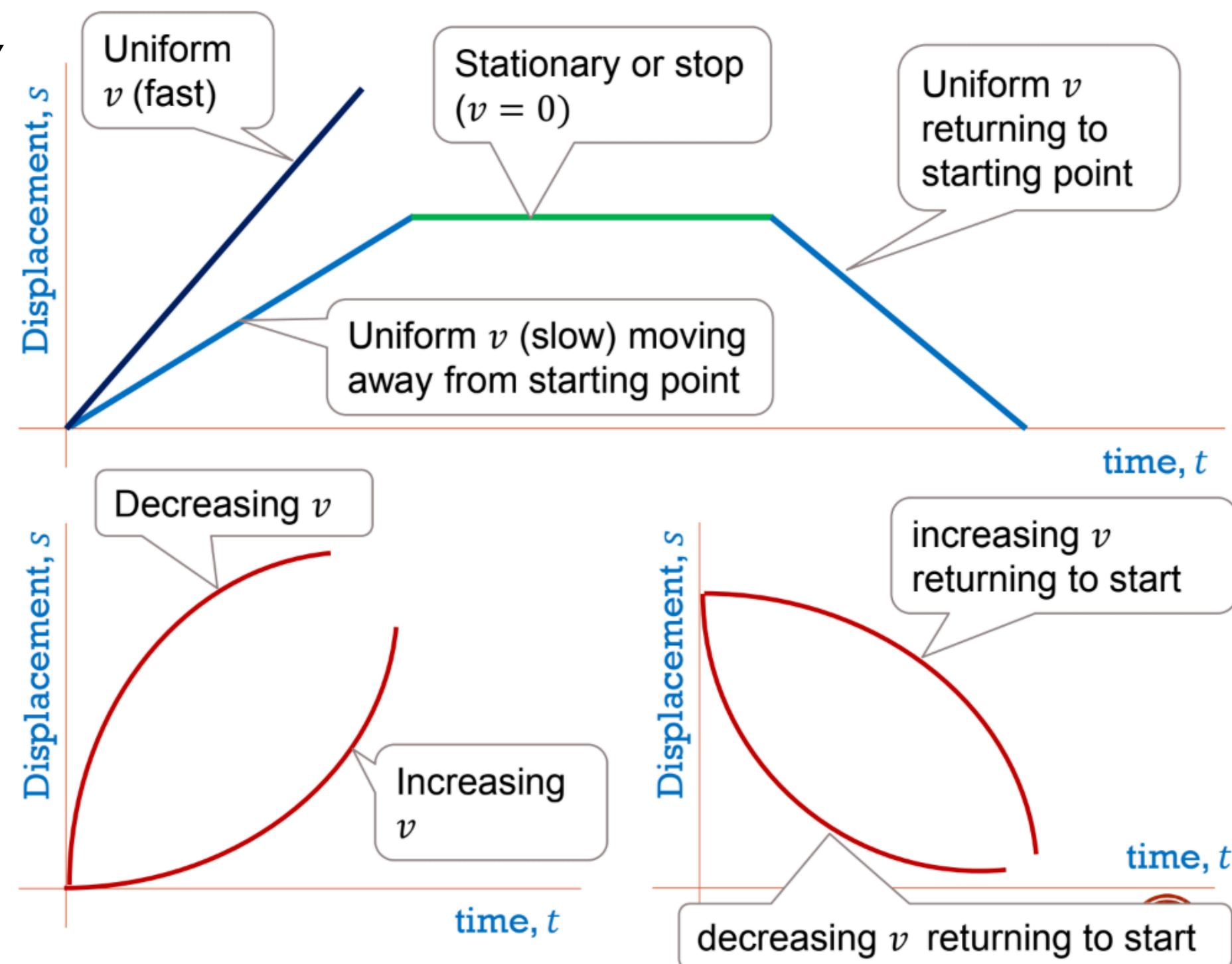
2.2 Motion Diagrams

2.2.1 Displacement-time (s – t)

The gradient of Displacement-Time graph is **Velocity, v**

If the gradient is:

- horizontal line
 - zero velocity / at rest / stationary / stop
- straight line slope (not horizontal)
 - uniform velocity / constant velocity
- Curve line
 - changing velocity with time

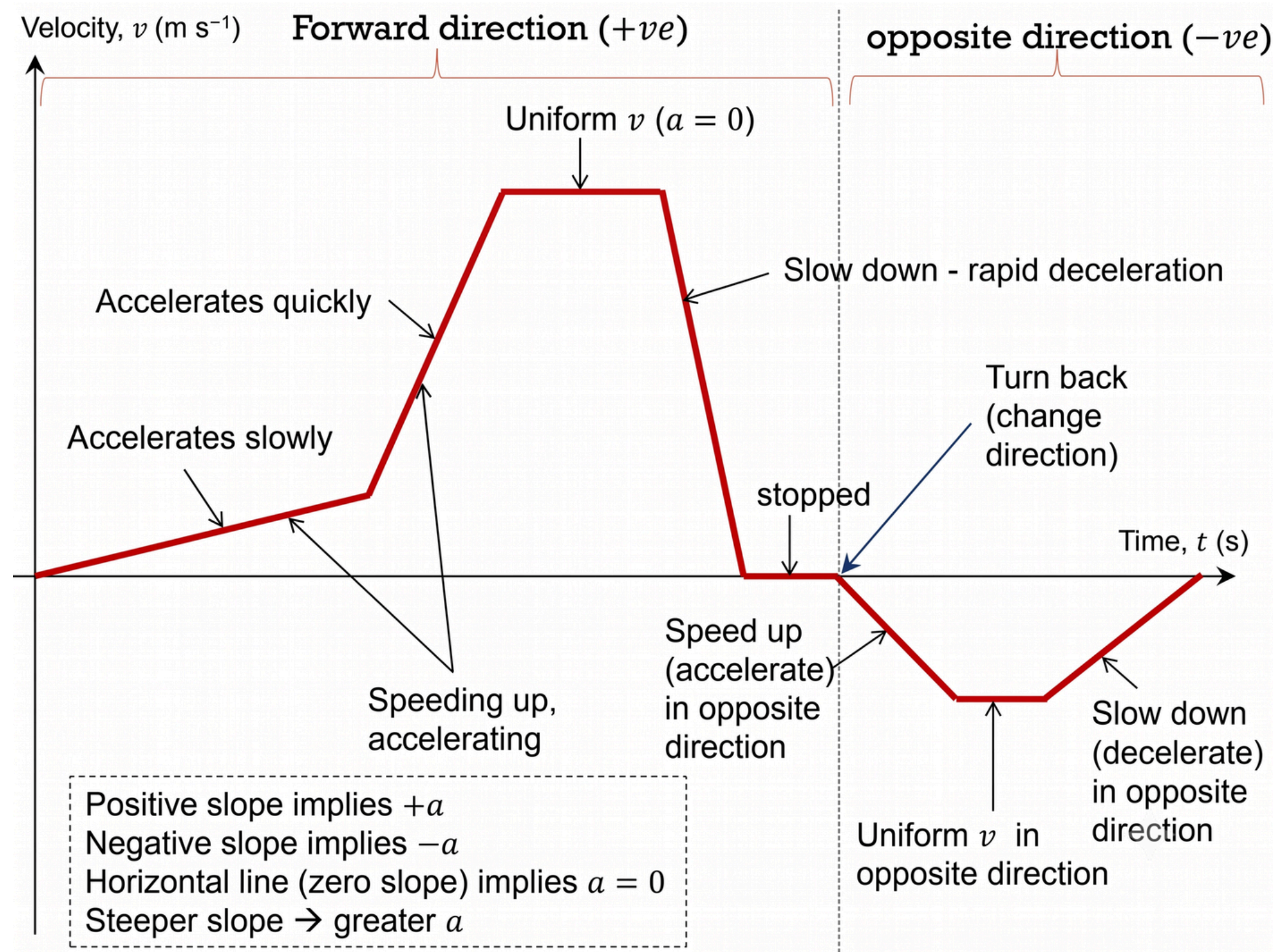
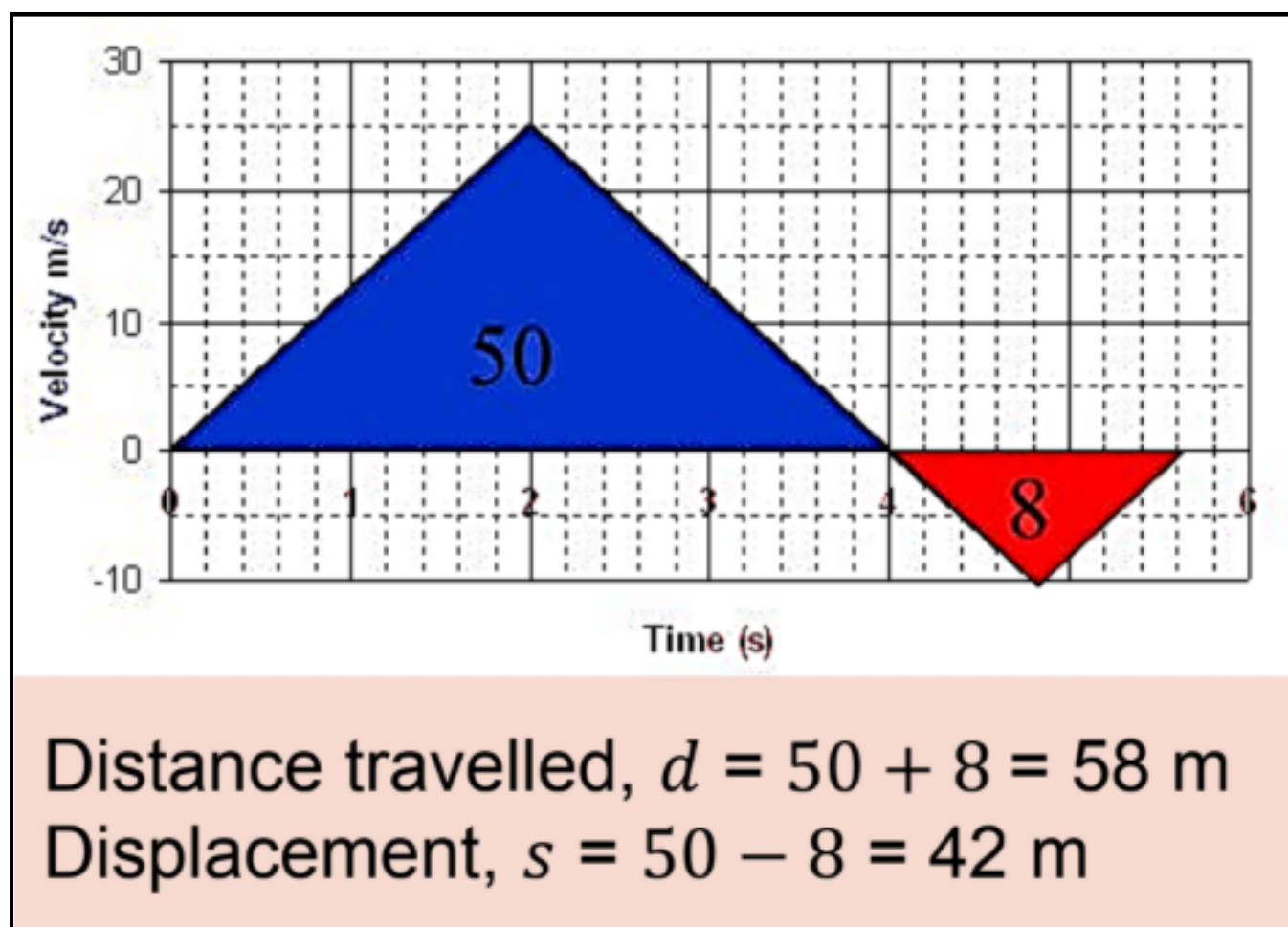


2.2 Motion Diagrams

2.2.2 Velocity-time ($v - t$)

The gradient of Displacement-Time graph is **acceleration, a** .

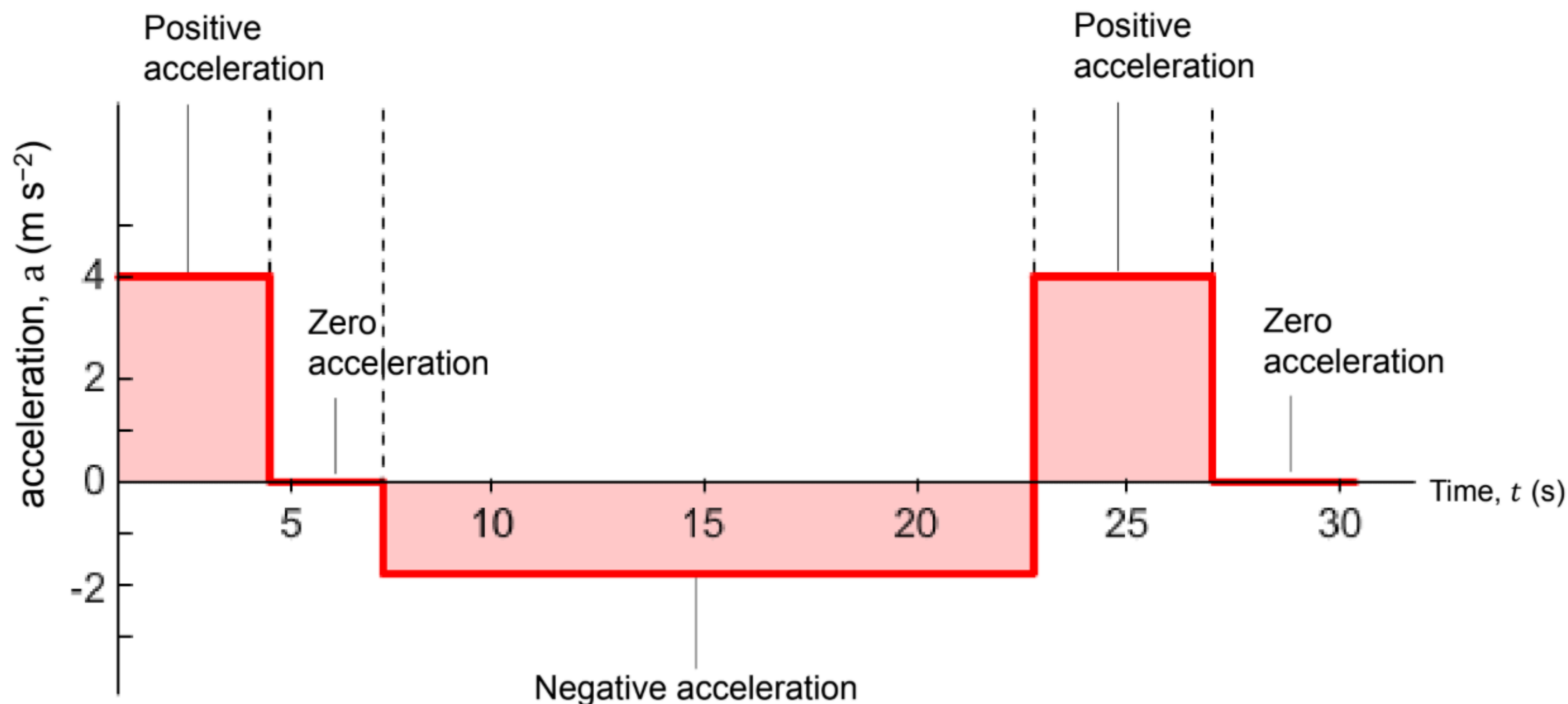
for velocity-time graph, the area under the graph is the **distance/displacement**



2.2 Motion Diagrams

2.2.3 Acceleration-time (a - t)

for acceleration-time graph, the area under the graph is the **change of velocity, Δv**



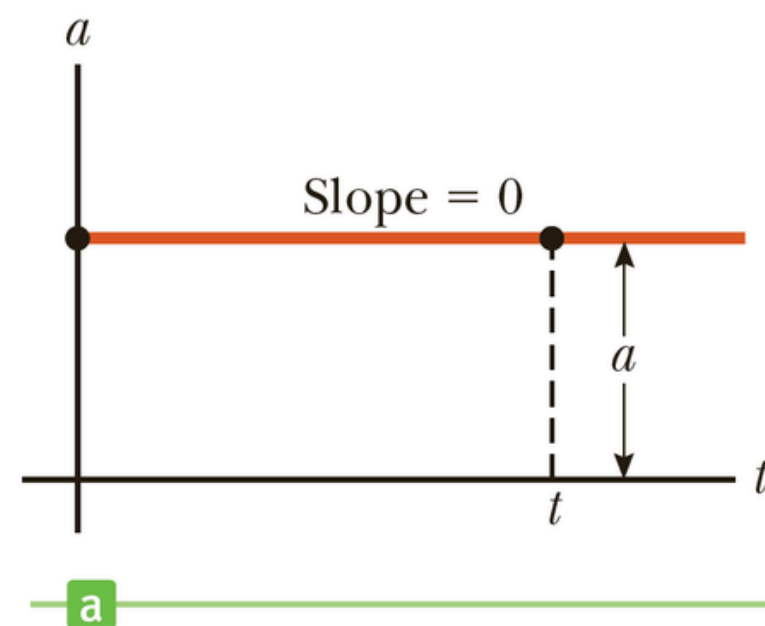
2.3 One-Dimensional Motion with Constant Acceleration

Constant acceleration is a crucial type of motion because it applies to numerous objects in nature, such as an object in free fall near Earth's surface (neglecting air resistance)

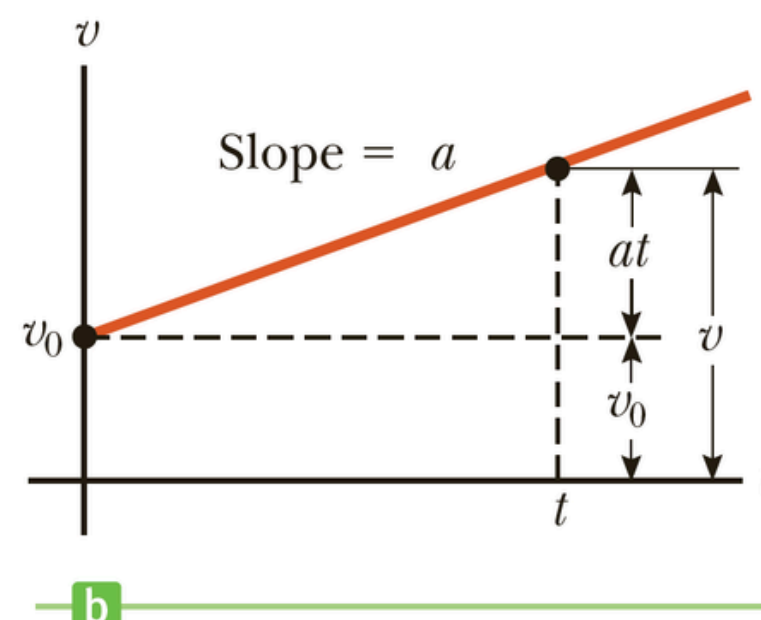
Key Characteristic:

When an object moves with constant acceleration, its instantaneous acceleration at any point matches its average acceleration over the entire interval.

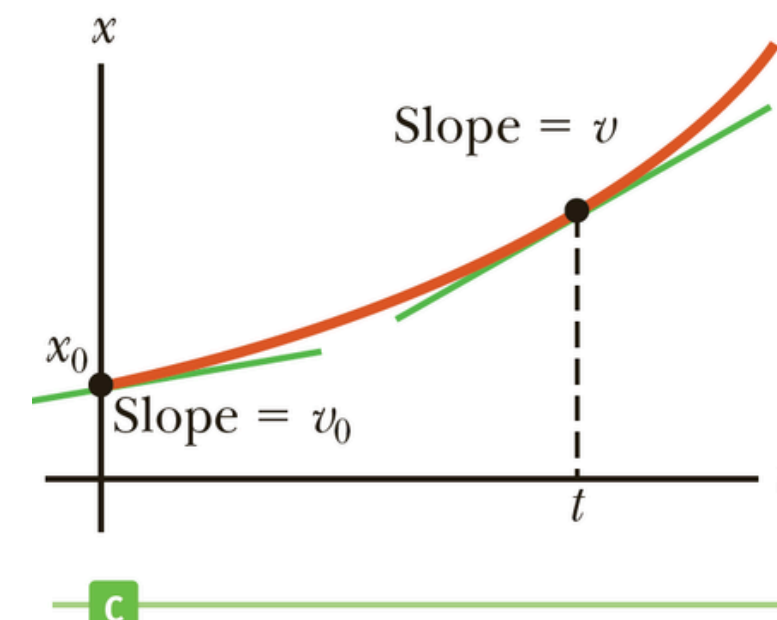
Graphical Representation of Constant Acceleration:



Gives a flat, horizontal line because the value remains unchanged over time (Slope = 0)



Since the velocity changes at a constant rate, this plot forms a straight line with a constant positive, zero, or negative slope. The slope represents the acceleration.



Forms a curved, parabolic path where the slope at any instant yields the instantaneous velocity (v)

2.3 One-Dimensional Motion with Constant Acceleration

Deriving the First Kinematic Equation (Velocity-Time)

Let define the primary variables for motion along the x-axis as follows:

- v_0 : Initial velocity of the object at time $t = 0$.
- v : Final velocity of the object at any arbitrary final time t .
- a : Constant acceleration maintained throughout the interval.
- t : The elapsed time or duration of motion (derived by assuming $t_i = 0$ and $t_f = t$).
- Δx : Displacement, defined as the change in position ($x - x_0$).

It is critical to note that all variables are vector quantities (except for time); therefore, a sign convention (+/−) must be strictly assigned based on the chosen coordinate system

2.3 One-Dimensional Motion with Constant Acceleration

Deriving the First Kinematic Equation (Velocity-Time)

Derivation 1: Velocity as a Function of Time

Starting with the definition of constant acceleration, $a = \frac{v_f - v_i}{t_f - t_i}$, and substituting our standard notation ($t_i = 0, t_f = t, v_i = v_0, v_f = v$), we obtain:

$$a = \frac{v - v_0}{t}$$

Rearranging for v yields:

$$v = v_0 + at$$

This equation shows that acceleration steadily modifies the initial velocity over time. The term **at** in the equation represents the total change in velocity added to the starting state.

2.3 One-Dimensional Motion with Constant Acceleration

Deriving the First Kinematic Equation (Velocity-Time)

Derivation 2: Displacement as a Function of Average Velocity

For constant acceleration, the velocity increases or decreases at a uniform rate. Consequently, the average velocity ($\bar{v}$) is the simple arithmetic mean of the initial and final velocities:

$$\bar{v} = \frac{v_0 + v}{2}$$

Substituting this into the definition of displacement ($\Delta x = \bar{v}t$), we derive:

$$\Delta x = \frac{1}{2}(v_0 + v)t$$

2.3 One-Dimensional Motion with Constant Acceleration

Deriving the First Kinematic Equation (Velocity-Time)

Derivation 3: Displacement as a Function of Time

By substituting the expression for v from the first kinematic equation into the displacement equation above, we eliminate the final velocity:

$$\Delta x = \frac{1}{2}(v_0 + [v_0 + at])t = v_0t + \frac{1}{2}at^2$$

Geometric Significance:

On a velocity-time graph, this represents the "area under the curve."

- The term v_0t represents the area of a rectangle, signifying the displacement that would occur if the object maintained its initial velocity
- The term $\frac{1}{2}at^2$ represents the area of a triangle, signifying the additional displacement contributed specifically by the acceleration.

2.3 One-Dimensional Motion with Constant Acceleration

Deriving the First Kinematic Equation (Velocity-Time)

Derivation 4: Displacement as a Function of Time

By solving the first equation for time ($t = \frac{v-v_0}{a}$) and substituting it into the displacement equation, we eliminate the time variable:

$$v^2 = v_0^2 + 2a\Delta x$$

Strategic Value: This is the preferred tool for scenarios where the duration of motion is unknown, allowing for a direct link between velocity and position.

2.3 One-Dimensional Motion with Constant Acceleration

Comparative Analysis of Kinematic Equations

Selecting the correct equation is the most critical step in professional problem-solving

- Use the "Missing Variable" as the hidden differentiator to guide your selection.

Equation	Information Given / Relationship	Missing Variable (The Differentiator)
$v = v_0 + at$	Velocity as a function of time	Displacement (Δx)
$\Delta x = \frac{1}{2}(v_0 + v)t$	Displacement as a function of average velocity	Acceleration (a)
$\Delta x = v_0t + \frac{1}{2}at^2$	Displacement as a function of time	Final Velocity (v)
$v^2 = v_0^2 + 2a\Delta x$	Velocity as a function of displacement	Time (t)

2.3 One-Dimensional Motion with Constant Acceleration

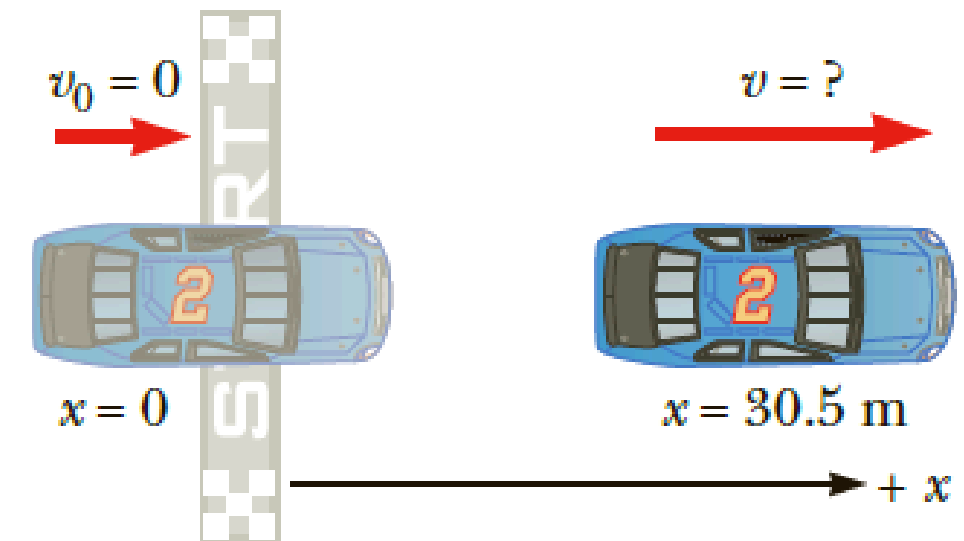
Strategic Methodology for Problem Solving

A disciplined approach is required to navigate the complexities of kinematics, particularly regarding sign conventions in deceleration or multi-phase scenarios

1. **Read and Analyze:** Thoroughly parse the problem statement to identify the physical phases of motion.
2. **Construct a Coordinate System:** Draw a diagram. Explicitly define the positive direction. Use arrows to indicate the directions of velocity and acceleration.
3. **Identify Quantities:** List known variables and circle the unknowns. Ensure all units are in SI (meters, seconds) and that signs (+/-) are assigned relative to your coordinate system.
4. **Select the Optimal Equation:** Reference the table above to find the equation that excludes the variable you neither have nor need.
5. **Solve for Unknowns:** Perform the algebraic manipulation. Note that this often results in solving one linear and one quadratic equation to find your two unknowns.

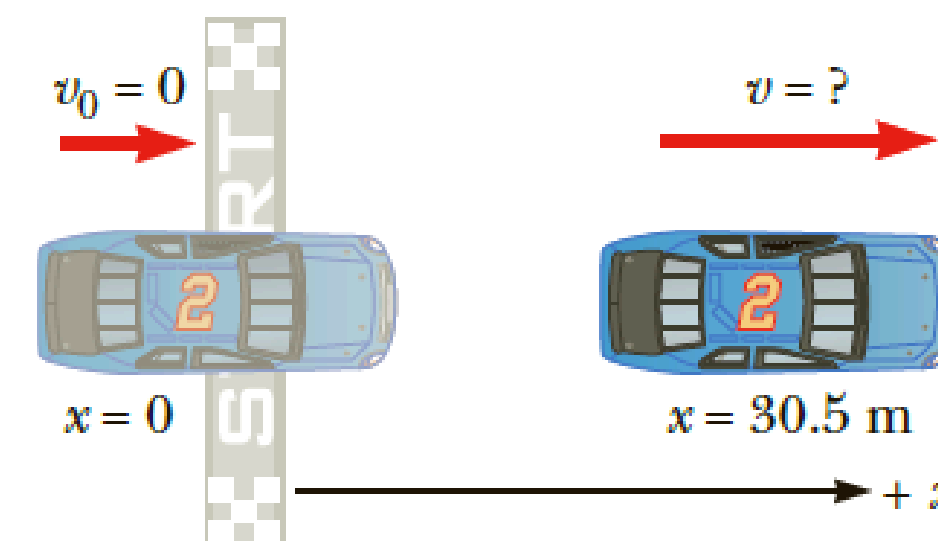
2.3 One-Dimensional Motion with Constant Acceleration

PROBLEM A race car starting from rest accelerates at a constant rate of 5.00 m/s^2 .
 (a) What is the velocity of the car after it has traveled $1.00 \times 10^2 \text{ ft}$? (b) How much time has elapsed? (c) Calculate the average velocity two different ways.



2.3 One-Dimensional Motion with Constant Acceleration

PROBLEM A race car starting from rest accelerates at a constant rate of 5.00 m/s^2 .
 (a) What is the velocity of the car after it has traveled $1.00 \times 10^2 \text{ ft}$? (b) How much time has elapsed? (c) Calculate the average velocity two different ways.



SOLUTION

(a) Convert units of Δx to SI, using the information in the inside front cover.

$$1.00 \times 10^2 \text{ ft} = (1.00 \times 10^2 \text{ ft}) \left(\frac{1 \text{ m}}{3.28 \text{ ft}} \right) = 30.5 \text{ m}$$

Write the kinematics equation for v^2 (step 4):

$$v^2 = v_0^2 + 2a \Delta x$$

Solve for v , taking the positive square root because the car moves to the right (step 5):

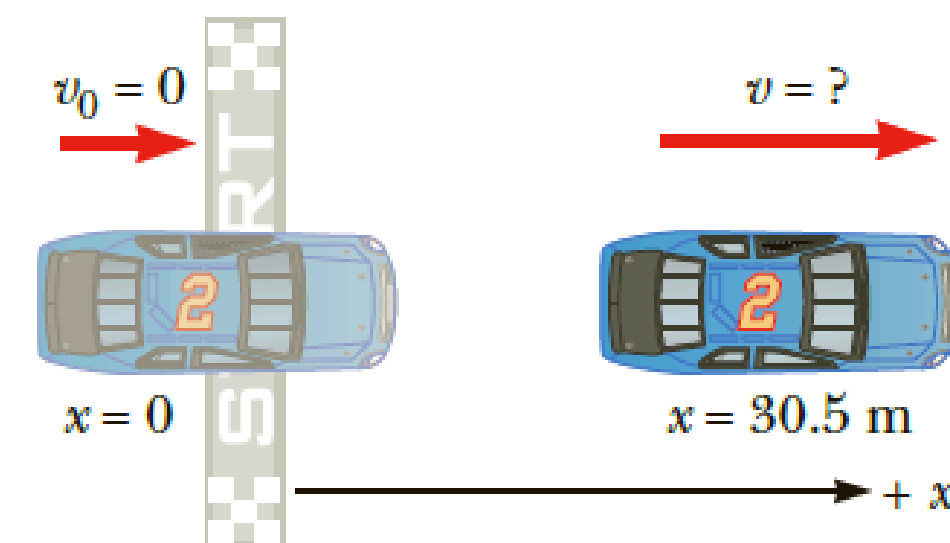
$$v = \sqrt{v_0^2 + 2a \Delta x}$$

Substitute $v_0 = 0$, $a = 5.00 \text{ m/s}^2$, and $\Delta x = 30.5 \text{ m}$:

$$\begin{aligned}
 v &= \sqrt{v_0^2 + 2a \Delta x} = \sqrt{(0)^2 + 2(5.00 \text{ m/s}^2)(30.5 \text{ m})} \\
 &= 17.5 \text{ m/s}
 \end{aligned}$$

2.3 One-Dimensional Motion with Constant Acceleration

PROBLEM A race car starting from rest accelerates at a constant rate of 5.00 m/s^2 .
 (a) What is the velocity of the car after it has traveled $1.00 \times 10^2 \text{ ft}$? (b) How much time has elapsed? (c) Calculate the average velocity two different ways.



SOLUTION

(b) How much time has elapsed?

Apply the first equation of Table 2.4:

Substitute values and solve for time t :

$$v = at + v_0$$

$$17.5 \text{ m/s} = (5.00 \text{ m/s}^2) t$$

$$t = \frac{17.5 \text{ m/s}}{5.00 \text{ m/s}^2} = 3.50 \text{ s}$$

(c) Calculate the average velocity in two different ways.

Apply the definition of average velocity, Equation 2.2:

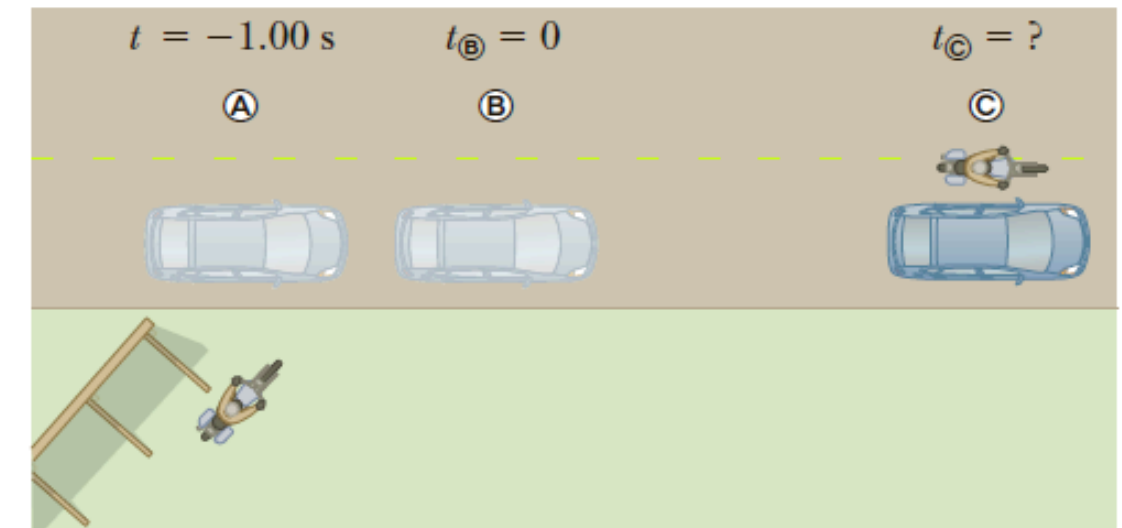
$$\bar{v} = \frac{x_f - x_i}{t_f - t_i} = \frac{30.5 \text{ m}}{3.50 \text{ s}} = 8.71 \text{ m/s}$$

Apply the definition of average velocity in Equation 2.7:

$$\bar{v} = \frac{v_0 + v}{2} = \frac{0 + 17.5 \text{ m/s}}{2} = 8.75 \text{ m/s}$$

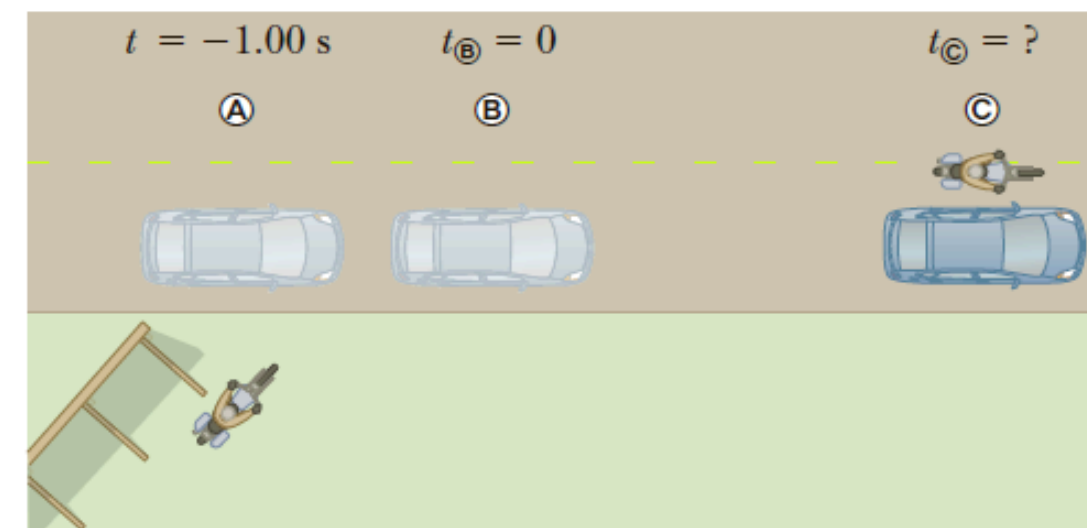
2.3 One-Dimensional Motion with Constant Acceleration

PROBLEM A car traveling at a constant speed of 24.0 m/s passes a trooper hidden behind a billboard, as in Figure 2.17. One second after the speeding car passes the billboard, the trooper sets off in chase with a constant acceleration of 3.00 m/s^2 . (a) How long does it take the trooper to overtake the speeding car? (b) How fast is the trooper going at that time?



2.3 One-Dimensional Motion with Constant Acceleration

PROBLEM A car traveling at a constant speed of 24.0 m/s passes a trooper hidden behind a billboard, as in Figure 2.17. One second after the speeding car passes the billboard, the trooper sets off in chase with a constant acceleration of 3.00 m/s². (a) How long does it take the trooper to overtake the speeding car? (b) How fast is the trooper going at that time?



SOLUTION

(a) How long does it take the trooper to overtake the car?

Write the equation for the car's displacement:

Take $x_0 = 24.0$ m, $v_0 = 24.0$ m/s, and $a_{\text{car}} = 0$. Solve for x_{car} :

Write the equation for the trooper's position, taking $x_0 = 0$, $v_0 = 0$, and $a_{\text{trooper}} = 3.00$ m/s²:

Set $x_{\text{trooper}} = x_{\text{car}}$, and solve the quadratic equation. (The quadratic formula appears in Appendix A, Equation A.8.) Only the positive root is meaningful.

$$\Delta x_{\text{car}} = x_{\text{car}} - x_0 = v_0 t + \frac{1}{2} a_{\text{car}} t^2$$

$$x_{\text{car}} = x_0 + v_0 t = 24.0 \text{ m} + (24.0 \text{ m/s}) t$$

$$x_{\text{trooper}} = \frac{1}{2} a_{\text{trooper}} t^2 = \frac{1}{2} (3.00 \text{ m/s}^2) t^2 = (1.50 \text{ m/s}^2) t^2$$

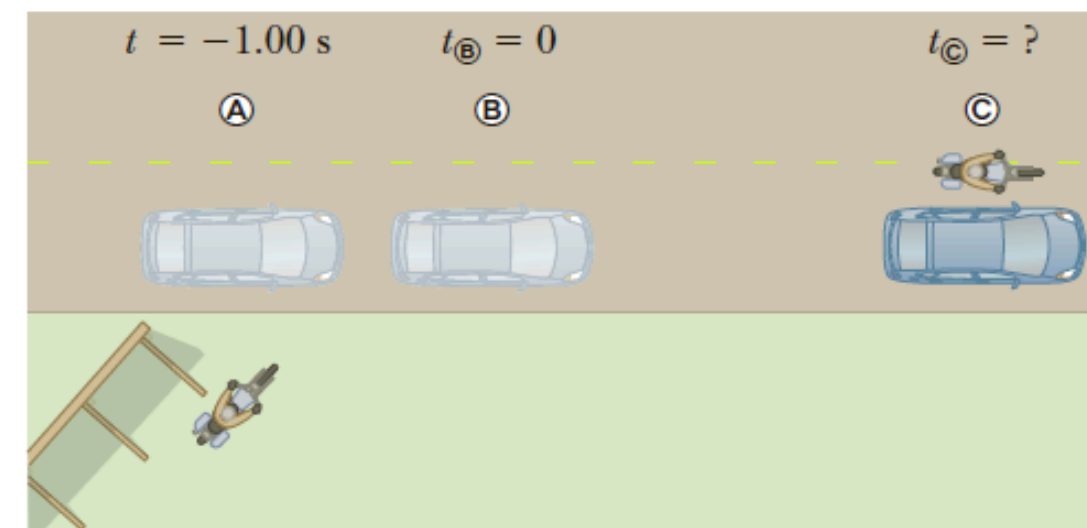
$$(1.50 \text{ m/s}^2) t^2 = 24.0 \text{ m} + (24.0 \text{ m/s}) t$$

$$(1.50 \text{ m/s}^2) t^2 - (24.0 \text{ m/s}) t - 24.0 \text{ m} = 0$$

$$t = 16.9 \text{ s}$$

2.3 One-Dimensional Motion with Constant Acceleration

PROBLEM A car traveling at a constant speed of 24.0 m/s passes a trooper hidden behind a billboard, as in Figure 2.17. One second after the speeding car passes the billboard, the trooper sets off in chase with a constant acceleration of 3.00 m/s². (a) How long does it take the trooper to overtake the speeding car? (b) How fast is the trooper going at that time?



SOLUTION

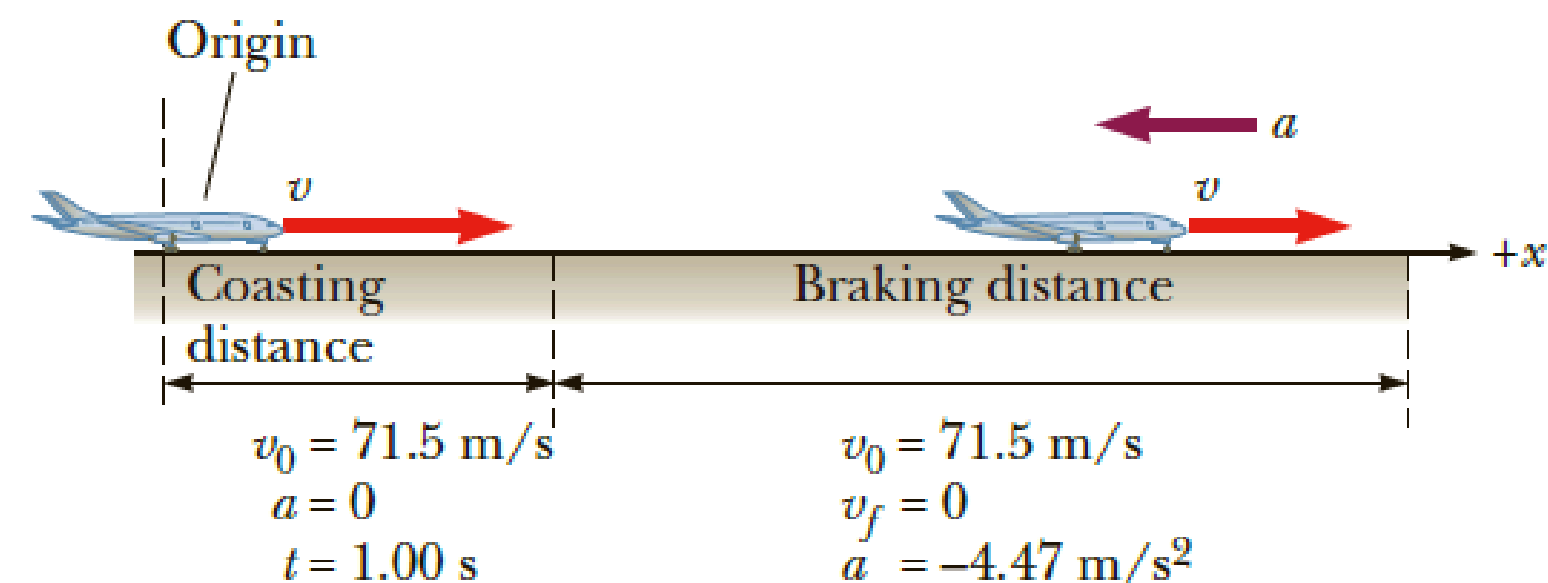
(b) Find the trooper's speed at that time.

Substitute the time into the trooper's velocity equation:

$$\begin{aligned}
 v_{\text{trooper}} &= v_0 + a_{\text{trooper}} t = 0 + (3.00 \text{ m/s}^2)(16.9 \text{ s}) \\
 &= 50.7 \text{ m/s}
 \end{aligned}$$

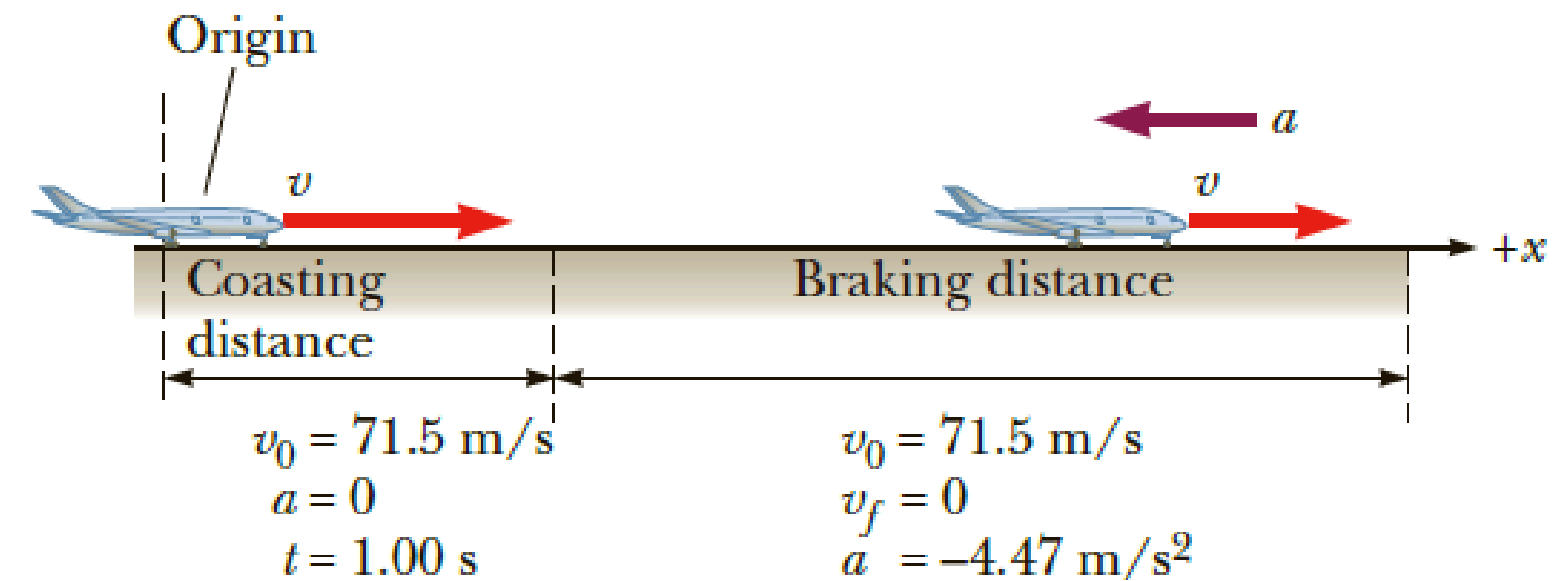
2.3 One-Dimensional Motion with Constant Acceleration

PROBLEM A typical jetliner lands at a speed of 1.60×10^2 mi/h and decelerates at the rate of $(10.0 \text{ mi/h})/\text{s}$. If the plane travels at a constant speed of 1.60×10^2 mi/h for 1.00 s after landing before applying the brakes, what is the total displacement of the aircraft between touchdown on the runway and coming to rest?



2.3 One-Dimensional Motion with Constant Acceleration

PROBLEM A typical jetliner lands at a speed of 1.60×10^2 mi/h and decelerates at the rate of $(10.0 \text{ mi/h})/\text{s}$. If the plane travels at a constant speed of 1.60×10^2 mi/h for 1.00 s after landing before applying the brakes, what is the total displacement of the aircraft between touchdown on the runway and coming to rest?



SOLUTION

Convert units of speed and acceleration to SI:

$$v_0 = (1.60 \times 10^2 \text{ mi/h}) \left(\frac{0.447 \text{ m/s}}{1.00 \text{ mi/h}} \right) = 71.5 \text{ m/s}$$

$$a = (-10.0 (\text{mi/h})/\text{s}) \left(\frac{0.447 \text{ m/s}}{1.00 \text{ mi/h}} \right) = -4.47 \text{ m/s}^2$$

Taking $a = 0$, $v_0 = 71.5 \text{ m/s}$, and $t = 1.00 \text{ s}$, find the displacement while the plane is coasting:

$$\Delta x_{\text{coasting}} = v_0 t + \frac{1}{2} a t^2 = (71.5 \text{ m/s})(1.00 \text{ s}) + 0 = 71.5 \text{ m}$$

Use the time-independent kinematic equation to find the displacement while the plane is braking.

$$v^2 = v_0^2 + 2 a \Delta x_{\text{braking}}$$

Take $a = -4.47 \text{ m/s}^2$ and $v_0 = 71.5 \text{ m/s}$. The negative sign on a means that the plane is slowing down.

$$\Delta x_{\text{braking}} = \frac{v^2 - v_0^2}{2a} = \frac{0 - (71.5 \text{ m/s})^2}{2.00(-4.47 \text{ m/s}^2)} = 572 \text{ m}$$

Sum the two results to find the total displacement:

$$\Delta x_{\text{coasting}} + \Delta x_{\text{braking}} = 71.5 \text{ m} + 572 \text{ m} = 644 \text{ m}$$

2.4 Freely Falling Objects

What is a Freely Falling Object?

A freely falling object is any object moving freely under the influence of gravity alone, completely independent of its initial motion.

This includes objects dropped from rest, thrown straight upward, or thrown straight downward.

The Key Condition: This idealized motion assumes that air resistance is negligible.



2.4 Freely Falling Objects

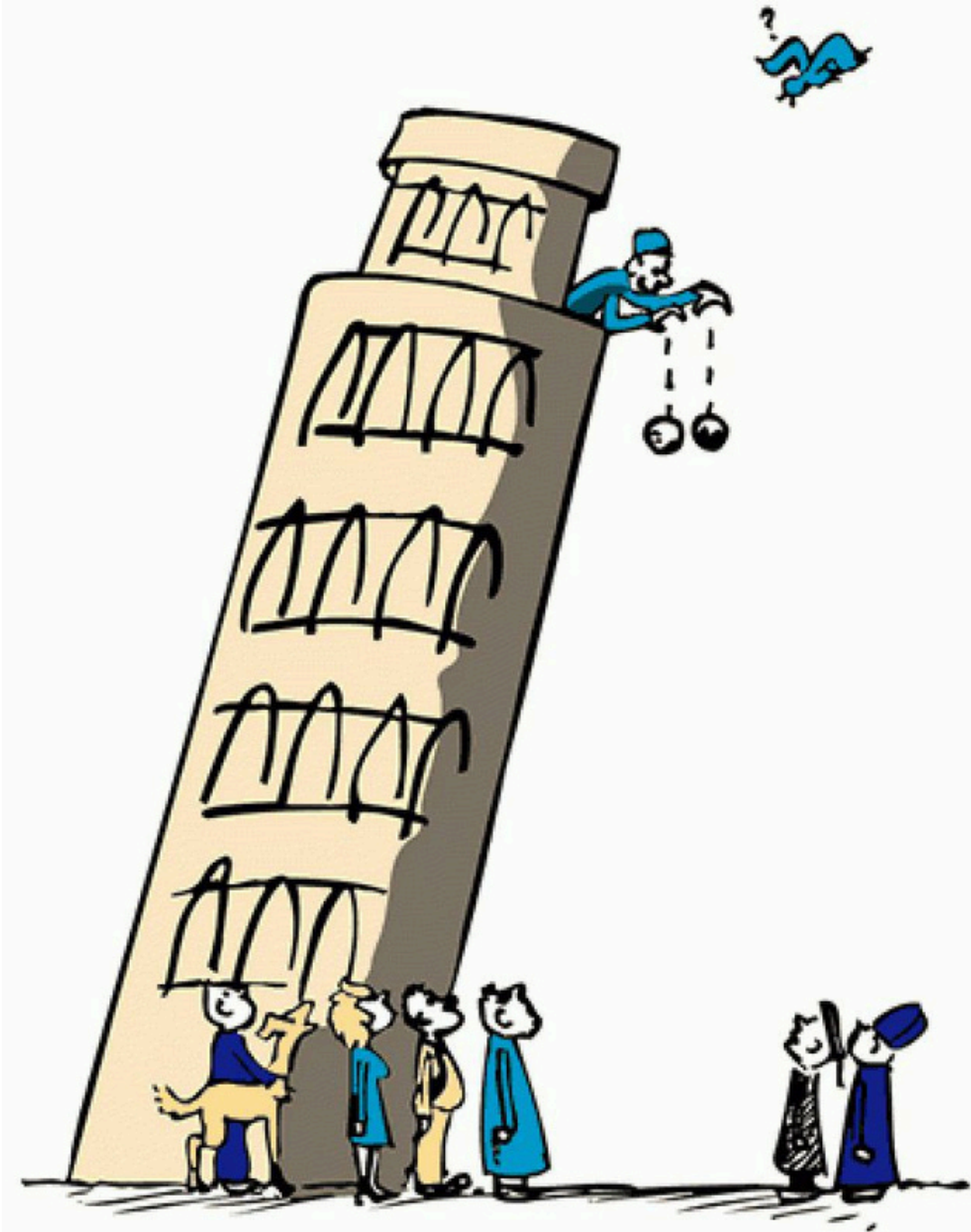
Aristotle vs. Galileo

Pre-1600 Perspective:

The philosopher Aristotle (384–322 BC) taught that heavier objects fall faster than lighter ones. This idea dominated until around 1600

Galileo Galilei (1564–1642): Formulated the actual laws governing free fall

- Legend says Galileo dropped two different weights simultaneously from the Leaning Tower of Pisa, observing that they hit the ground at the same time. (Pisa Legend)
[note: ofcourse this is unlikely to happen]



Aspect	Aristotelian View (Pre-1600)	Galilean Experimental Findings
Premise of Falling	Heavier objects fall faster than lighter ones.	All objects fall with the same constant acceleration.
Methodology	Philosophical teaching and casual observation.	Systematic experimentation and measurement.
Role of Mass	Mass determines the speed of the fall.	Mass has no effect on acceleration when air resistance is neglected.

2.4 Freely Falling Objects

Real-World Evidence & The Acceleration of Gravity

The Hammer vs. Feather Experiment:

On Earth, a hammer hits the floor before a feather because air drag affects the lighter feather much more.

On August 2, 1971, Apollo 15 astronaut David Scott dropped a hammer and a feather on the Moon (where there is no air).

- They fell with the exact same acceleration and hit the surface simultaneously.
- <https://www.youtube.com/watch?v=oYEgdZ3iEKA>



The magnitude of free-fall acceleration is denoted by **g**.

- standard value at surface of the Earth: $g = 9.81 \text{ m s}^{-2}$ (standard for calculation)
- for quick estimation: $g = 10 \text{ m s}^{-2}$

The value of **g** decreases as altitude increases, and it varies slightly with latitude.

2.4 Freely Falling Objects

Applying Kinematics to One-Dimensional Free Fall

If **air resistance is neglected** over short vertical distances, free fall is identical to 1D motion under **constant acceleration**.

Therefore, standard 1D kinematics equations can be fully applied.

$$v = v_0 + at$$

$$\Delta x = v_0 t + \frac{1}{2}at^2$$

$$\Delta x = \frac{1}{2}(v_0 + v)t$$

$$v^2 = v_0^2 + 2a\Delta x$$

Coordinate System Conventions:

- Define "up" as the positive vertical direction (positive y-direction).
- Use y-axis as the position variable
- Because gravity pulls down, acceleration is always negative: $a = -g = -9.80 \text{ m s}^{-2}$.

2.4 Freely Falling Objects

Applying Kinematics to One-Dimensional Free Fall

Because $a = -g$ for free fall and we use y instead of x , we have the following

$$\Delta x = v_i t + \frac{1}{2} a t^2$$



$$\Delta y = v_i t - \frac{1}{2} g t^2$$

$$v_f = v_i + a t$$

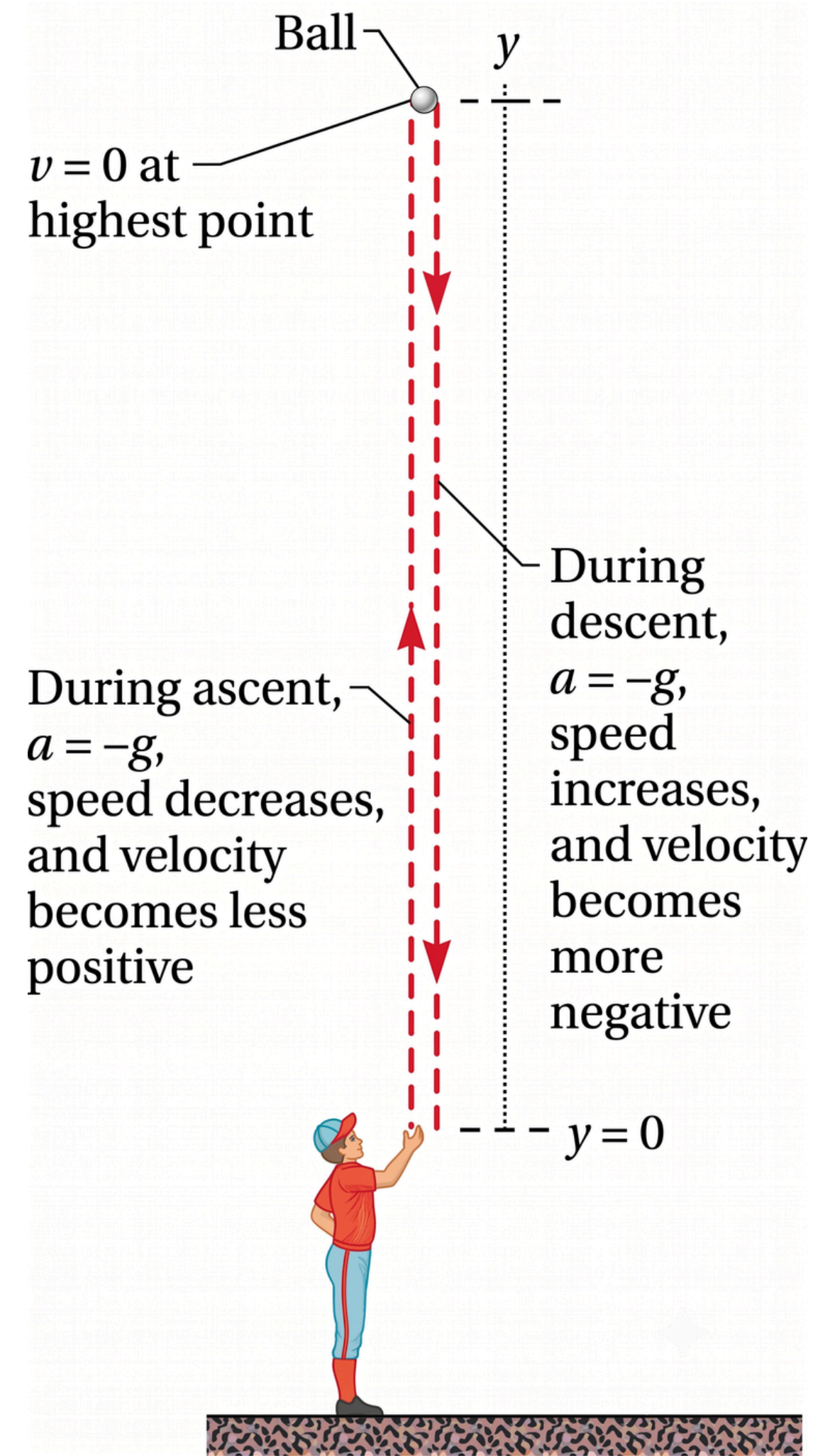


$$v_f = v_i - g t$$

$$v_f^2 = v_i^2 + 2 a \Delta x$$



$$v_f^2 = v_i^2 - 2 g \Delta y$$



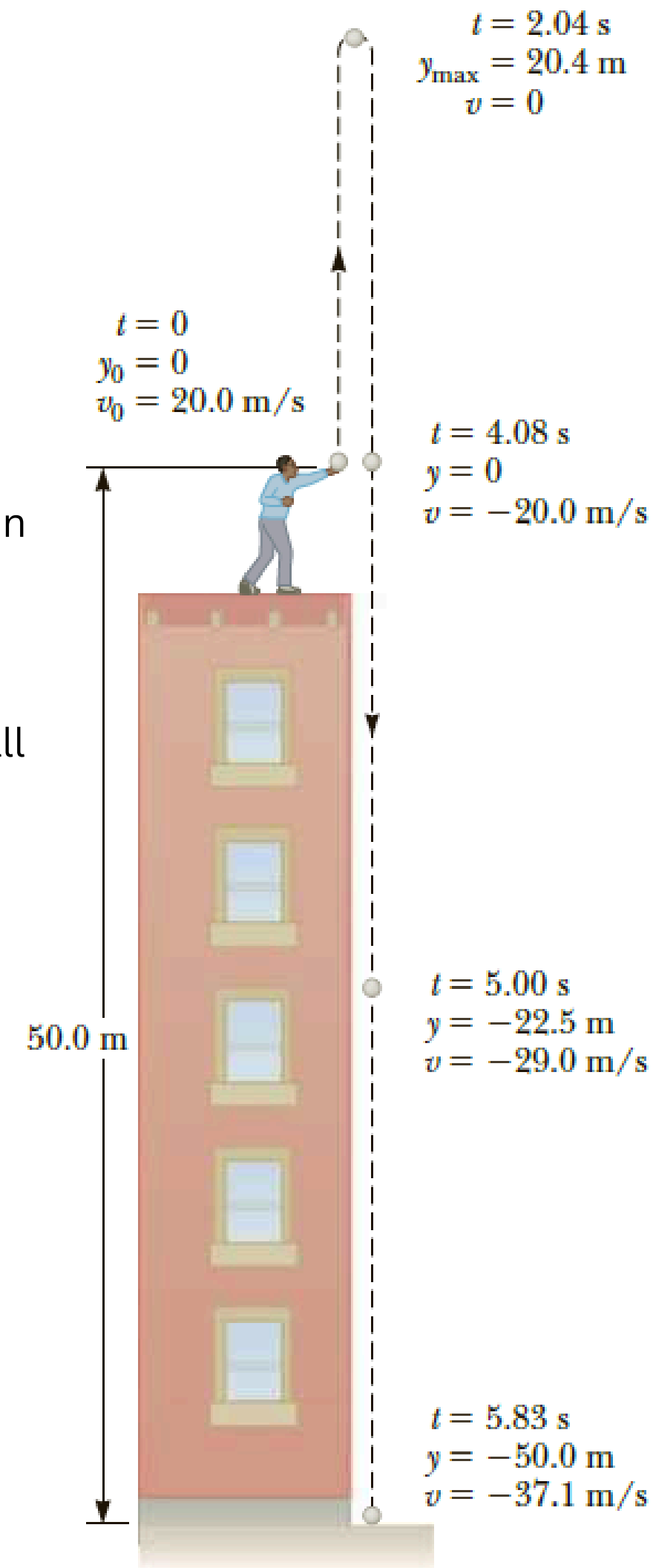
2.4 Freely Falling Objects

Example

A ball is thrown from the top of a building with an initial velocity of 20.0 m/s straight upward, at an initial height of 50.0 m above the ground. The ball just misses the edge of the roof on its way down, as shown in the figure. Determine

- the time needed for the ball to reach its maximum height,
- the maximum height,
- the time needed for the ball to return to the height from which it was thrown and the velocity of the ball at that instant,
- the time needed for the ball to reach the ground, and
- the velocity and position of the ball at $t = 5.00$ s.

Neglect air drag.



2.4 Freely Falling Objects

Example

A ball is thrown from the top of a building with an initial velocity of 20.0 m/s straight upward, at an initial height of 50.0 m above the ground. The ball just misses the edge of the roof on its way down, as shown in the figure. Determine

(a) the time needed for the ball to reach its maximum height,

SOLUTION

(a) Find the time when the ball reaches its maximum height.

Write the velocity and position kinematic equations:

$$v = at + v_0$$

$$\Delta y = y - y_0 = v_0 t + \frac{1}{2}at^2$$

Substitute $a = -9.80 \text{ m/s}^2$, $v_0 = 20.0 \text{ m/s}$, and $y_0 = 0$ into the preceding two equations:

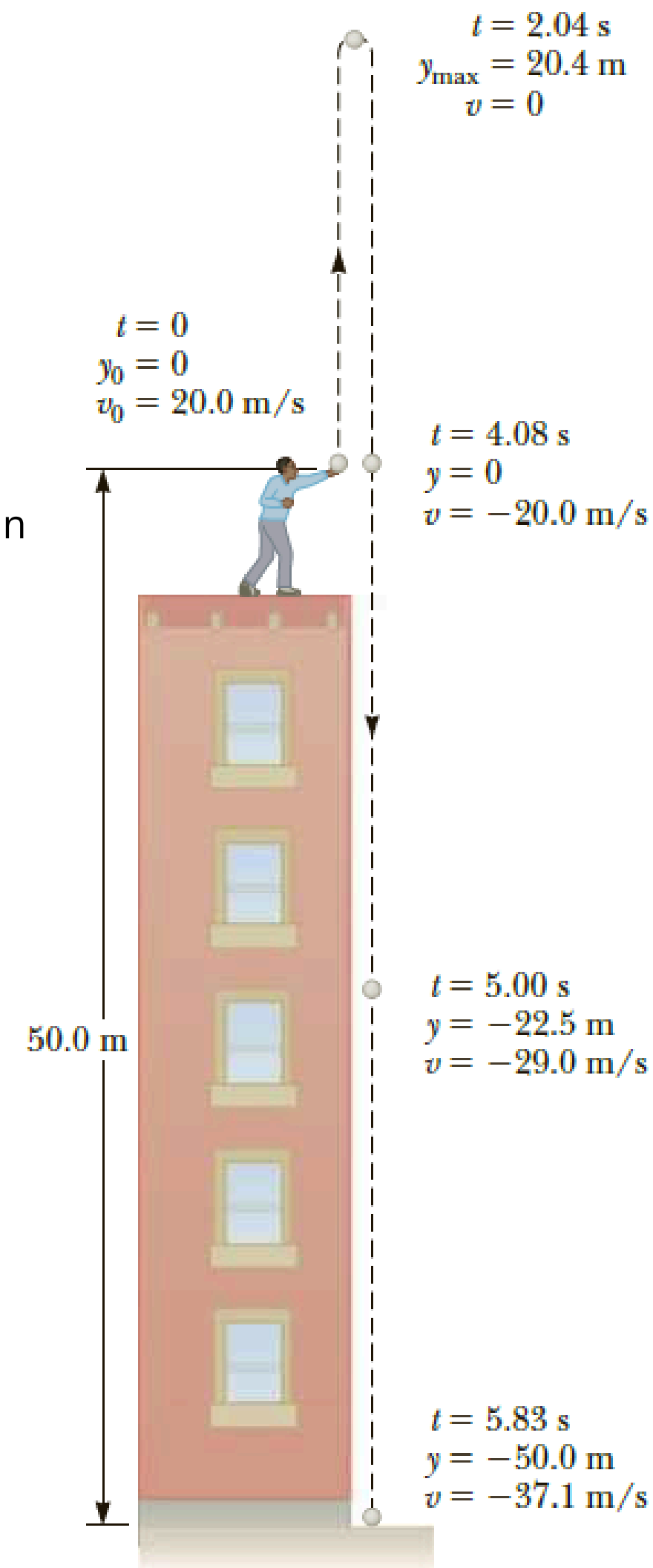
$$(1) \quad v = (-9.80 \text{ m/s}^2)t + 20.0 \text{ m/s}$$

$$(2) \quad y = (20.0 \text{ m/s})t - (4.90 \text{ m/s}^2)t^2$$

Substitute $v = 0$, the velocity at maximum height, into Equation (1) and solve for time:

$$0 = (-9.80 \text{ m/s}^2)t + 20.0 \text{ m/s}$$

$$t = \frac{-20.0 \text{ m/s}}{-9.80 \text{ m/s}^2} = 2.04 \text{ s}$$



2.4 Freely Falling Objects

Example

A ball is thrown from the top of a building with an initial velocity of 20.0 m/s straight upward, at an initial height of 50.0 m above the ground. The ball just misses the edge of the roof on its way down, as shown in the figure. Determine

- (b) the maximum height,
- (c) the time needed for the ball to return to the height from which it was thrown and the velocity of the ball at that instant,

SOLUTION

- (b) Determine the ball's maximum height.

Substitute the time $t = 2.04$ s into Equation (2):

$$y_{\max} = (20.0 \text{ m/s})(2.04 \text{ s}) - (4.90 \text{ m/s}^2)(2.04 \text{ s})^2 = 20.4 \text{ m}$$

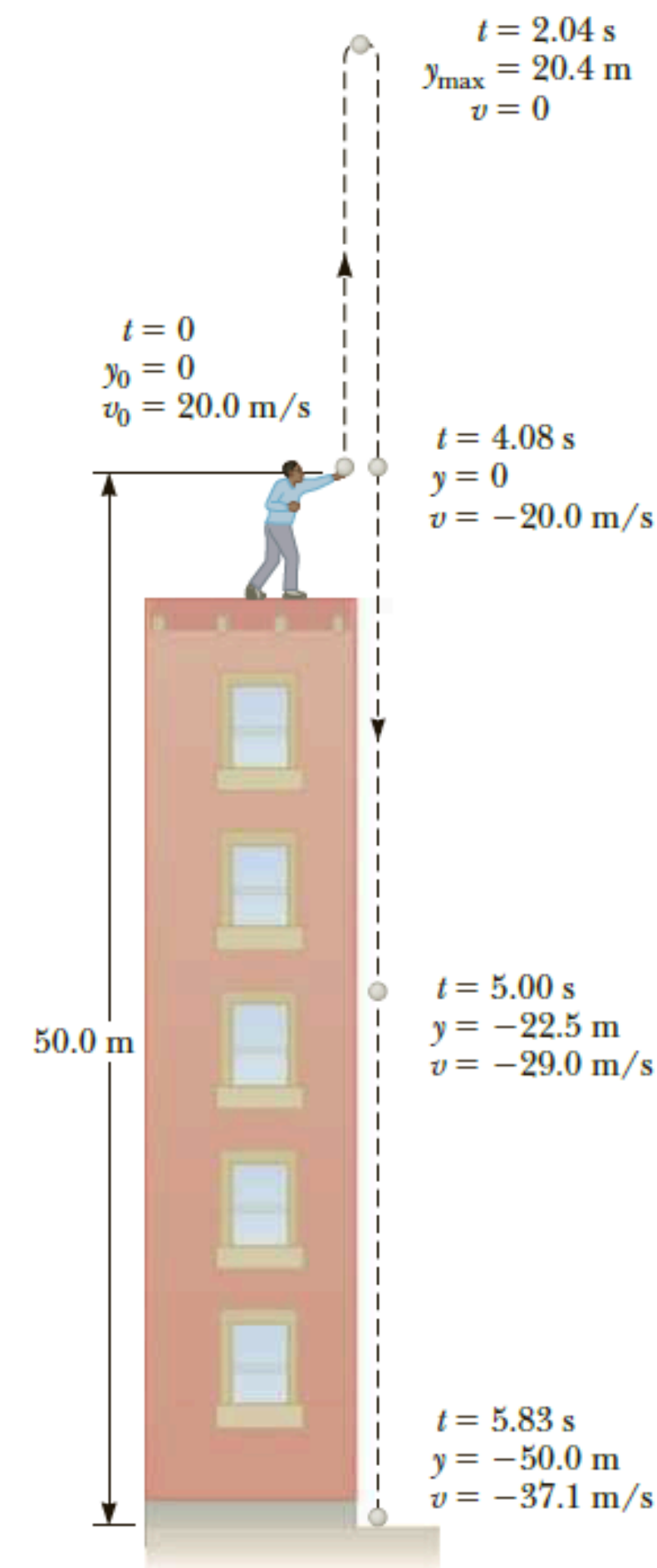
- (c) Find the time the ball takes to return to its initial position, and find the velocity of the ball at that time.

Set $y = 0$ in Equation (2) and solve for t :

$$\begin{aligned}
 0 &= (20.0 \text{ m/s})t - (4.90 \text{ m/s}^2)t^2 \\
 &= t(20.0 \text{ m/s} - (4.90 \text{ m/s}^2)t) \\
 t &= 4.08 \text{ s}
 \end{aligned}$$

Substitute the time into Equation (1) to get the velocity:

$$v = 20.0 \text{ m/s} + (-9.80 \text{ m/s}^2)(4.08 \text{ s}) = -20.0 \text{ m/s}$$



2.4 Freely Falling Objects

Example

A ball is thrown from the top of a building with an initial velocity of 20.0 m/s straight upward, at an initial height of 50.0 m above the ground. The ball just misses the edge of the roof on its way down, as shown in the figure. Determine

- (d) the time needed for the ball to reach the ground, and
- (e) the velocity and position of the ball at $t = 5.00$ s.

SOLUTION

- (d) Find the time required for the ball to reach the ground.

In Equation (2), set $y = -50.0$ m:

$$-50.0 \text{ m} = (20.0 \text{ m/s})t - (4.90 \text{ m/s}^2)t^2$$

$$t = 5.83 \text{ s}$$

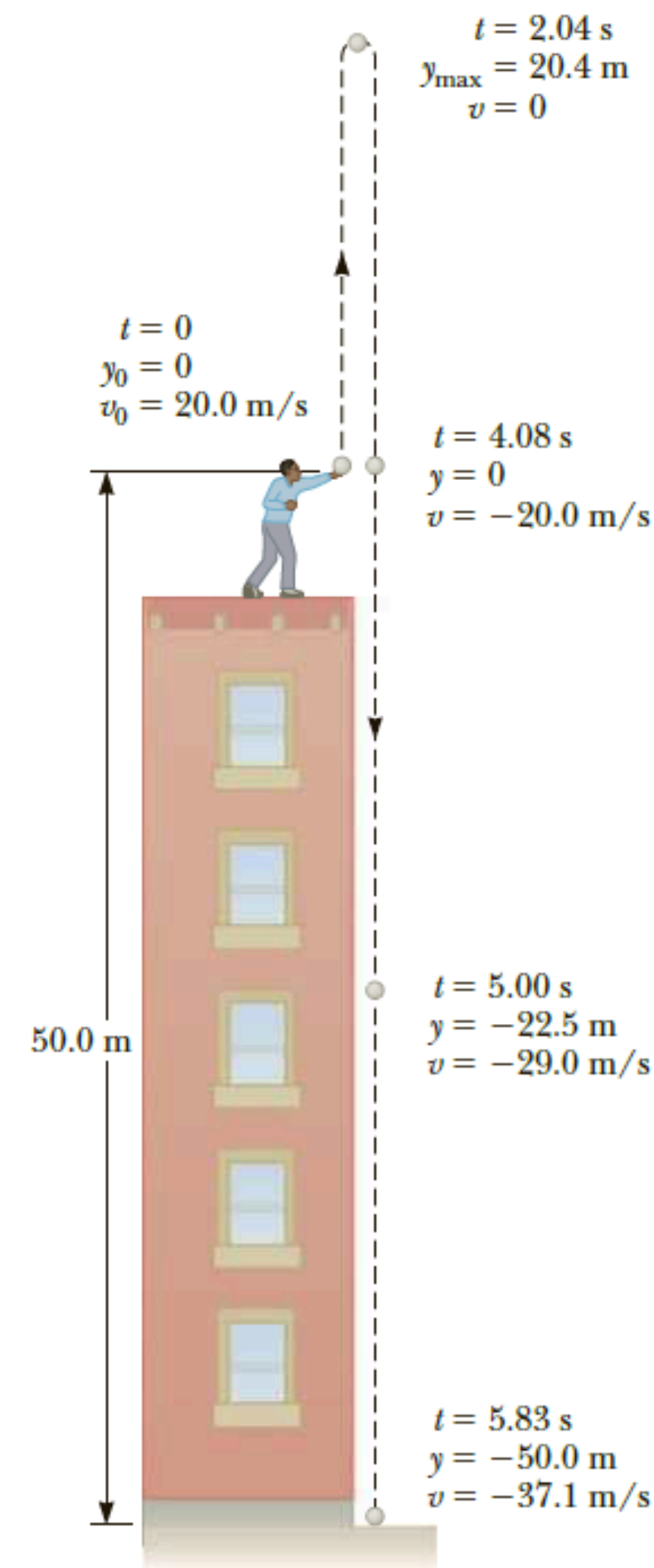
Apply the quadratic formula and take the positive root:

- (e) Find the velocity and position of the ball at $t = 5.00$ s.

Substitute values into Equations (1) and (2):

$$v = (-9.80 \text{ m/s}^2)(5.00 \text{ s}) + 20.0 \text{ m/s} = -29.0 \text{ m/s}$$

$$y = (20.0 \text{ m/s})(5.00 \text{ s}) - (4.90 \text{ m/s}^2)(5.00 \text{ s})^2 = -22.5 \text{ m}$$



2.4 Freely Falling Objects

Example

A projectile is launched straight up at 60.0 m/s from a height of 80.0 m, at the edge of a sheer cliff. The projectile falls, just missing the cliff and hitting the ground below. Find:

- (a) the maximum height of the projectile above the point of firing,
- (b) the time it takes to hit the ground at the base of the cliff
- (c) its velocity at impact.

2.4 Freely Falling Objects

Example

A projectile is launched straight up at 60.0 m/s from a height of 80.0 m, at the edge of a sheer cliff. The projectile falls, just missing the cliff and hitting the ground below. Find:

- (a) the maximum height of the projectile above the point of firing,
- (b) the time it takes to hit the ground at the base of the cliff
- (c) its velocity at impact.

SOLUTION

(a) Find the maximum height of the projectile above the point of firing

At the maximum height, the projectile momentarily comes to rest, which means its final velocity at that point is $v = 0$.

We use the time-independent kinematic equation:

$$v^2 = v_0^2 + 2a\Delta y$$

Substitute the known values into the equation:

$$0 = (60.0 \text{ m/s})^2 + 2(-9.80 \text{ m/s}^2)y_{\text{max}}$$

$$0 = 3600 - 19.6y_{\text{max}}$$

Rearrange and solve for y_{max} :

$$19.6y_{\text{max}} = 3600$$

$$y_{\text{max}} = \frac{3600}{19.6} \approx 184 \text{ m}$$

2.4 Freely Falling Objects

Example

A projectile is launched straight up at 60.0 m/s from a height of 80.0 m, at the edge of a sheer cliff. The projectile falls, just missing the cliff and hitting the ground below. Find:

- (a) the maximum height of the projectile above the point of firing,
- (b) the time it takes to hit the ground at the base of the cliff
- (c) its velocity at impact.

SOLUTION

(b) Find the time it takes to hit the ground at the base of the cliff

When the projectile hits the ground at the base of the cliff, its position is $y = -80.0$ m.

We use the displacement kinematic equation:

$$y = v_0t + \frac{1}{2}at^2$$

Substitute the known values:

$$-80.0 = (60.0)t + \frac{1}{2}(-9.80)t^2$$

$$-80.0 = 60.0t - 4.90t^2$$

Rearrange this into a standard quadratic equation ($at^2 + bt + c = 0$):

$$4.90t^2 - 60.0t - 80.0 = 0$$

Apply the quadratic formula, $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

$$t = \frac{-(-60.0) \pm \sqrt{(-60.0)^2 - 4(4.90)(-80.0)}}{2(4.90)}$$

$$t = \frac{60.0 \pm 71.89}{9.80}$$

Since time cannot be negative, we take the positive root:

$$t = \frac{60.0 + 71.89}{9.80} = \frac{131.89}{9.80} \approx \mathbf{13.5 \text{ s}}$$

2.4 Freely Falling Objects

Example

A projectile is launched straight up at 60.0 m/s from a height of 80.0 m, at the edge of a sheer cliff. The projectile falls, just missing the cliff and hitting the ground below. Find:

- (a) the maximum height of the projectile above the point of firing,
- (b) the time it takes to hit the ground at the base of the cliff
- (c) its velocity at impact.

SOLUTION

(c) Find its velocity at impact

To find the final velocity (v) just before hitting the ground, we can use the velocity kinematic equation:

$$v = v_0 + at$$

Following the rounding method of the textbook, we substitute the rounded time calculated in part (b) ($t = 13.5$ s):

$$v = 60.0 \text{ m/s} + (-9.80 \text{ m/s}^2)(13.5 \text{ s})$$

$$v = 60.0 - 132.3$$

$$v = -\mathbf{72.3 \text{ m/s}}$$

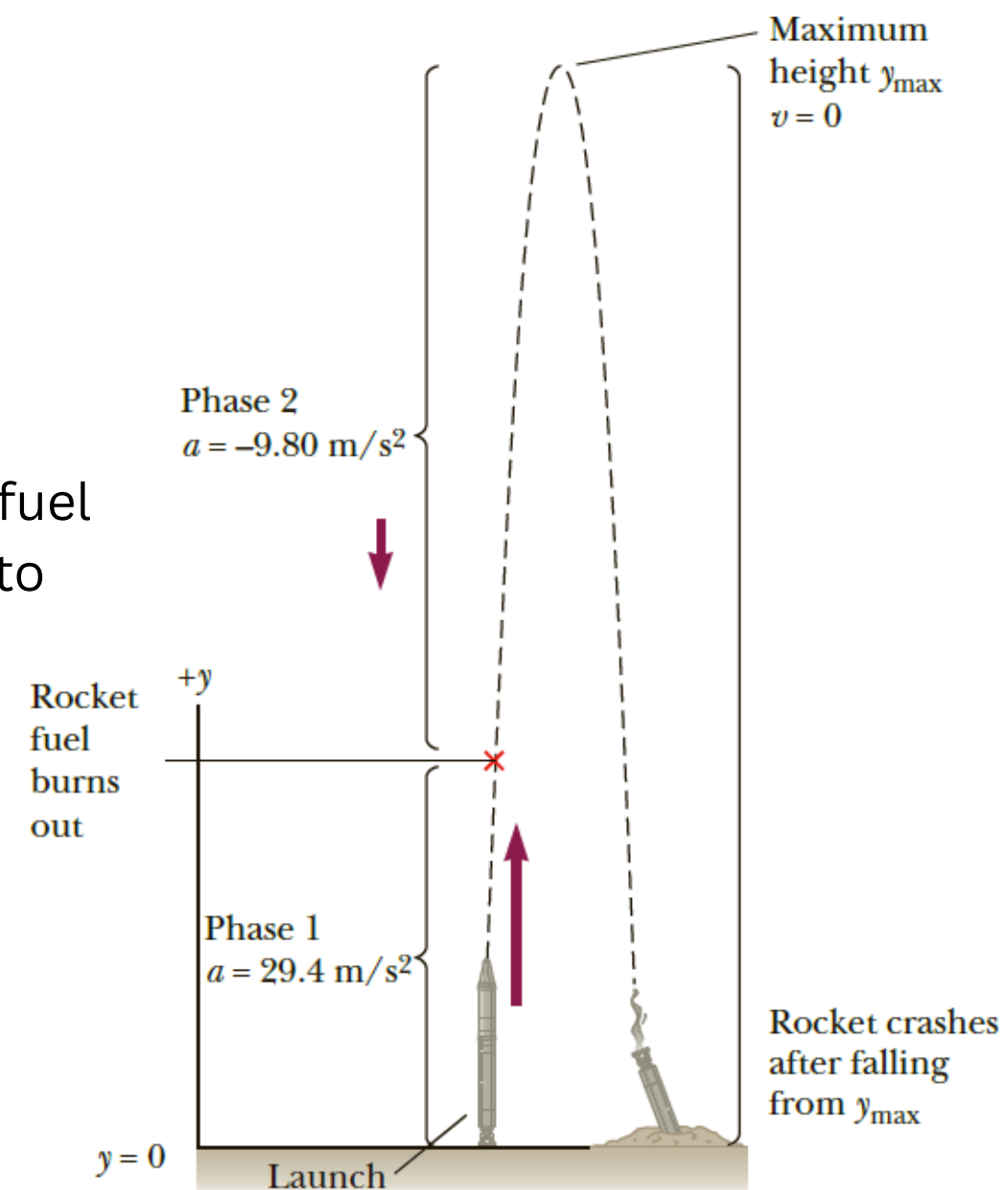
Note: The negative sign indicates that the projectile is traveling downward at the moment of impact.

2.4 Freely Falling Objects

Example

A rocket moves straight upward, starting from rest with an acceleration of 129.4 m/s^2 . It runs out of fuel at the end of 4.00 s and continues to coast upward, reaching a maximum height before falling back to Earth.

- Find the rocket's velocity and position at the end of 4.00 s .
- Find the maximum height the rocket reaches.
- Find the velocity the instant before the rocket crashes on the ground.



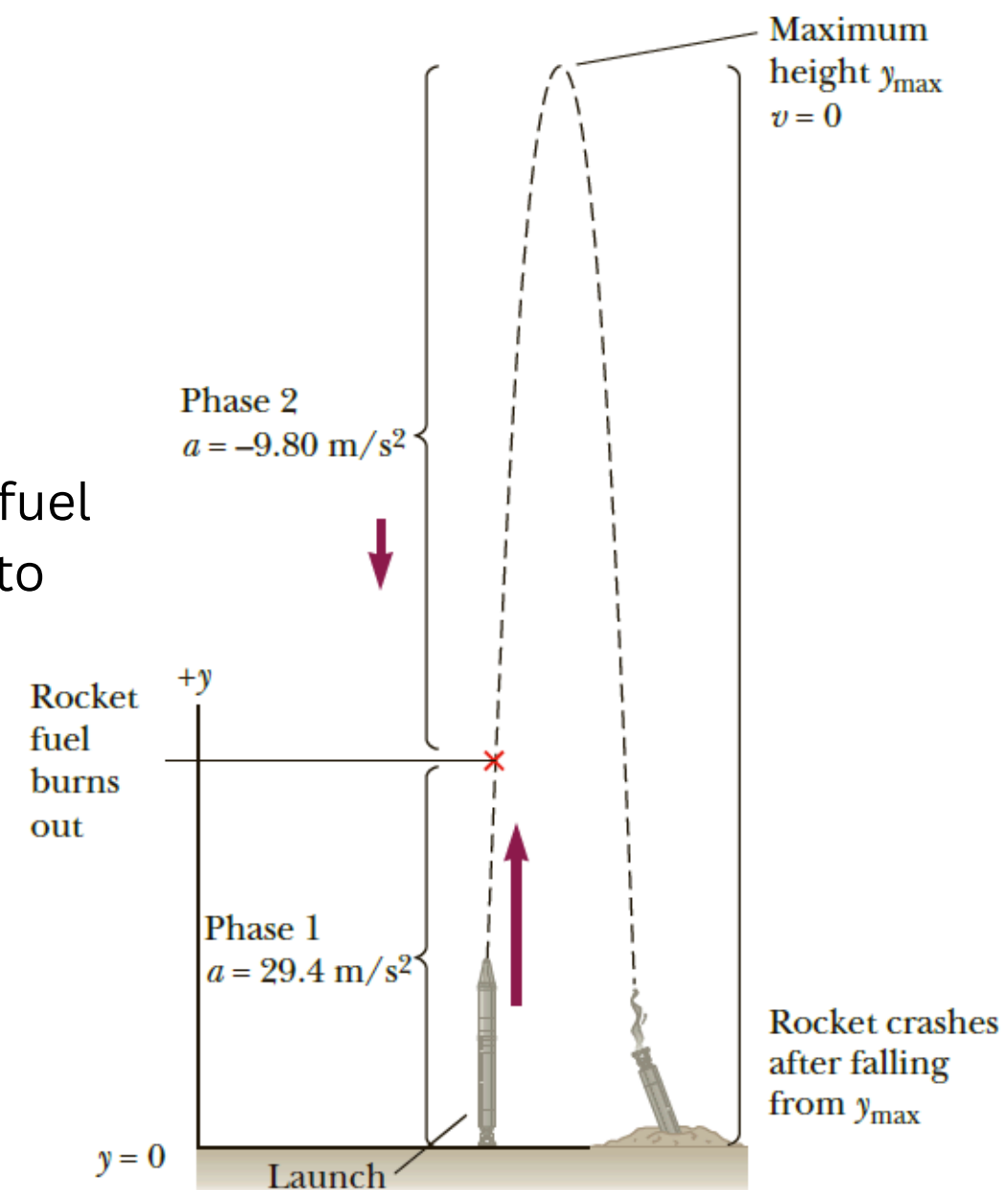
2.4 Freely Falling Objects

Example

A rocket moves straight upward, starting from rest with an acceleration of 129.4 m/s^2 . It runs out of fuel at the end of 4.00 s and continues to coast upward, reaching a maximum height before falling back to Earth.

- Find the rocket's velocity and position at the end of 4.00 s .
- Find the maximum height the rocket reaches.
- Find the velocity the instant before the rocket crashes on the ground.

SOLUTION



2.4 Freely Falling Objects

Example

A rocket moves straight upward, starting from rest with an acceleration of 129.4 m/s^2 . It runs out of fuel at the end of 4.00 s and continues to coast upward, reaching a maximum height before falling back to Earth.

- Find the rocket's velocity and position at the end of 4.00 s .
- Find the maximum height the rocket reaches.
- Find the velocity the instant before the rocket crashes on the ground.

SOLUTION

(a) Phase 1: Find the rocket's velocity and position after 4.00 s .

Write the velocity and position kinematic equations:

Adapt these equations to phase 1, substituting $a = 29.4 \text{ m/s}^2$, $v_0 = 0$, and $y_0 = 0$:

Substitute $t = 4.00 \text{ s}$ into Equations (3) and (4) to find the rocket's velocity v and position y at the time of burnout. These will be called v_b and y_b , respectively.

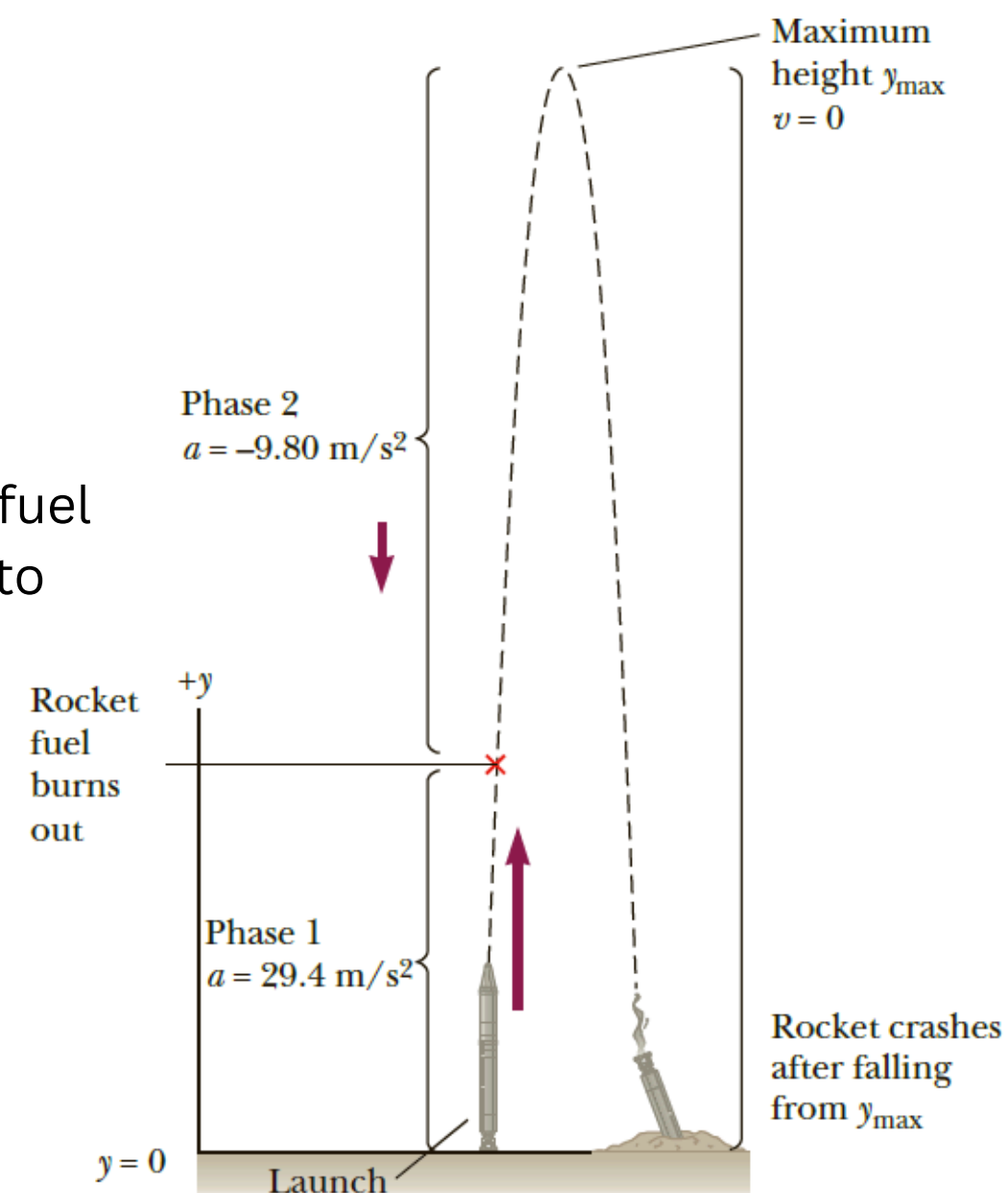
$$(1) \quad v = v_0 + at$$

$$(2) \quad \Delta y = y - y_0 = v_0 t + \frac{1}{2} at^2$$

$$(3) \quad v = (29.4 \text{ m/s}^2)t$$

$$(4) \quad y = \frac{1}{2}(29.4 \text{ m/s}^2)t^2 = (14.7 \text{ m/s}^2)t^2$$

$$v_b = 118 \text{ m/s} \quad \text{and} \quad y_b = 235 \text{ m}$$



2.4 Freely Falling Objects

Example

A rocket moves straight upward, starting from rest with an acceleration of 129.4 m/s^2 . It runs out of fuel at the end of 4.00 s and continues to coast upward, reaching a maximum height before falling back to Earth.

- Find the rocket's velocity and position at the end of 4.00 s .
- Find the maximum height the rocket reaches.
- Find the velocity the instant before the rocket crashes on the ground.

SOLUTION

(b) Phase 2: Find the maximum height the rocket attains.

Adapt Equations (1) and (2) to phase 2, substituting $a = -9.8 \text{ m/s}^2$, $v_0 = v_b = 118 \text{ m/s}$, and $y_0 = y_b = 235 \text{ m}$:

Substitute $v = 0$ (the rocket's velocity at maximum height) in Equation (5) to get the time it takes the rocket to reach its maximum height:

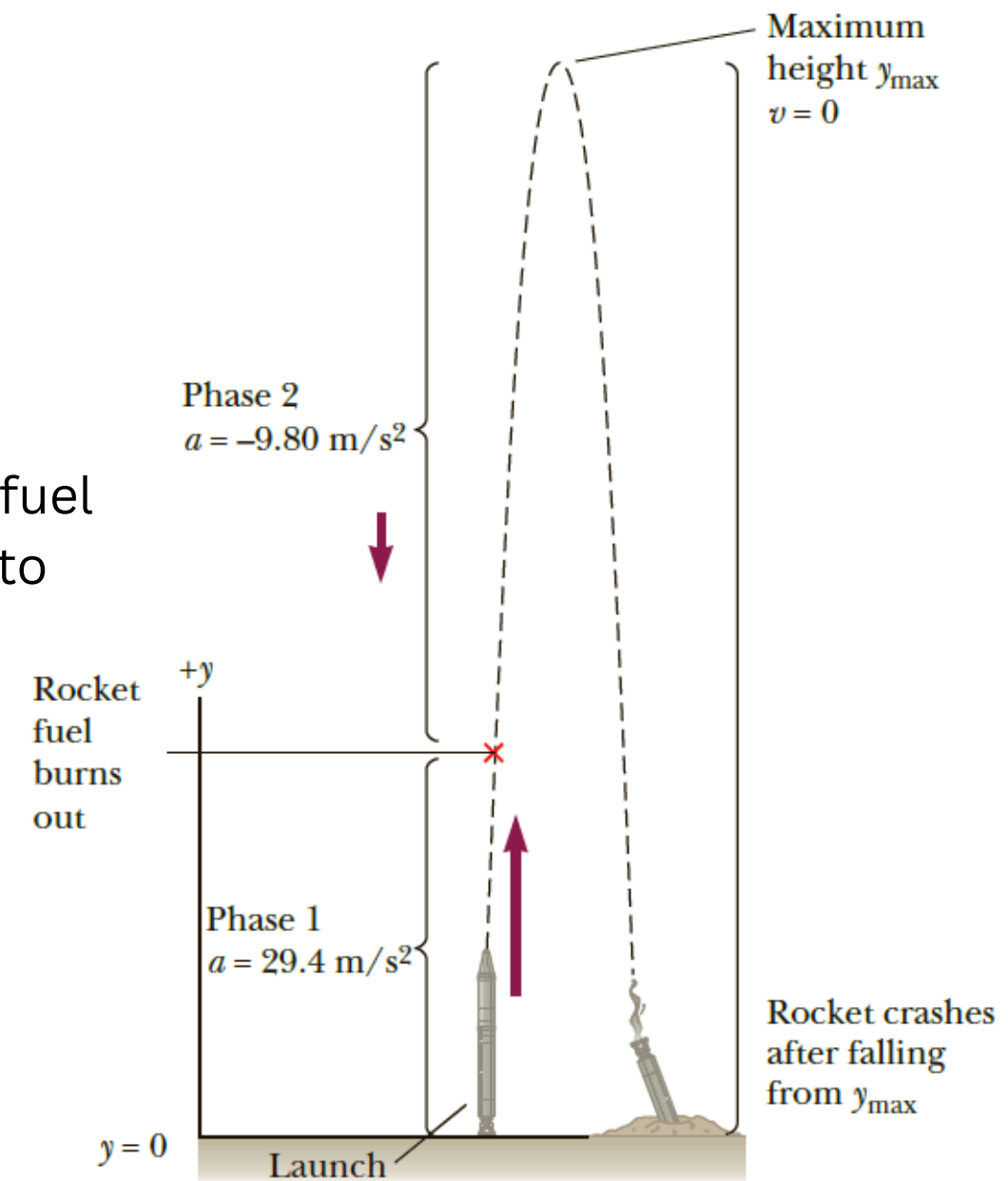
Substitute $t = 12.0 \text{ s}$ into Equation (6) to find the rocket's maximum height:

$$(5) \quad v = (-9.8 \text{ m/s}^2)t + 118 \text{ m/s}$$

$$(6) \quad y = 235 \text{ m} + (118 \text{ m/s})t - (4.90 \text{ m/s}^2)t^2$$

$$0 = (-9.8 \text{ m/s}^2)t + 118 \text{ m/s} \rightarrow t = \frac{118 \text{ m/s}}{9.80 \text{ m/s}^2} = 12.0 \text{ s}$$

$$\begin{aligned} y_{\text{max}} &= 235 \text{ m} + (118 \text{ m/s})(12.0 \text{ s}) - (4.90 \text{ m/s}^2)(12.0 \text{ s})^2 \\ &= 945 \text{ m} \end{aligned}$$



2.4 Freely Falling Objects

Example

A rocket moves straight upward, starting from rest with an acceleration of 129.4 m/s^2 . It runs out of fuel at the end of 4.00 s and continues to coast upward, reaching a maximum height before falling back to Earth.

- Find the rocket's velocity and position at the end of 4.00 s .
- Find the maximum height the rocket reaches.
- Find the velocity the instant before the rocket crashes on the ground.

SOLUTION

(c) Phase 2: Find the velocity of the rocket just prior to impact.

Find the time to impact by setting $y = 0$ in Equation (6) and using the quadratic formula:

Substitute this value of t into Equation (5):

$$0 = 235 \text{ m} + (118 \text{ m/s})t - (4.90 \text{ m/s}^2)t^2$$

$$t = 25.9 \text{ s}$$

$$v = (-9.80 \text{ m/s}^2)(25.9 \text{ s}) + 118 \text{ m/s} = -136 \text{ m/s}$$

